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Affective Pathways to Sustainability

Emotions, Meaning Making, and Transformation Through Non-Profit-Led Environmental and Sustainability Education

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Abstract

This thesis is an explorative study of how primary school students experience place-based Environmental and Sustainability Education programs, with a particular focus on affective engagement and its influence on meaning making. Two non-profit environmental education programs that collaborate with local actors and address locally embedded environmental challenges form the context of this study. Grounded in Sipos et al.'s (2008) Head-Heart-Hands framework for Transformative Sustainability Learning, Dewey's philosophy of experience, and Pekrun's Control-Value Theory of emotions, this study examines the conditions under which affective experiences arise in these programs and how students respond to these experiences.

It argues that emotions are not secondary outcomes but a central mechanism in sustainability learning. Data from field observations and student interviews identify aesthetic, place-based, and social experiences as key elements that shape students' emotional experiences. These experiences, in turn, support students' understanding of and engagement with environmental issues and contribute to their meaning making, wherein local actors play a central role. This leads to emerging identity transformations, in which environmental care is valued and pursued.

The findings further show that the affective, cognitive, and practical domains mutually reinforce each other, and that by emphasizing either the cognitive or practical dimension alongside the affective dimension, distinct but complementary pathways towards transformative learning for sustainability are shaped. The study underscores the importance of pedagogical design in shaping students' emotional engagement and indicates that activating emotions for sustainability learning can be meaningfully cultivated already in primary school, offering concrete insights for program designers and classroom educators.

Keywords: Environmental and Sustainability Education, Transformative Learning for Sustainability, Experiential Learning, Meaning Making, Emotions

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From Stien: To my lovely people at home, thank you for being my safe haven to which I will always return. And to all the people whom I crossed paths with during my travels – you have been my transformative experience leading to this thesis. Thank you en met liefs x

From Louisa: To my family and friends, both here and gone, your love and care have meant everything. Every flight, laugh, and attempt to read any of this work will stay with me, always. And, to my sweet Little Bear.

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Louisa Kennedy & Stien Naets

Authors' Contributions

This thesis is the result of a collaborative effort, and both authors take joint responsibility for the work in its entirety. While individual sections were written by one author or the other, the selection of theoretical perspectives, interpretation of the literature, and integration of ideas were discussed and expanded on collaboratively throughout the research process.

Stien Naets was primarily responsible for drafting the Introduction and the following parts of the Literature Review: Transformative Learning for Sustainability (Section 3.1), Head–Heart–Hands Framework (Section 3.1.1), Emotions as Judgements of Control and Value (Section 3.4.2), and Negative Emotions in ESE: a Driver or Barrier? (Section 3.4.3). Stien also produced the initial draft of the Methods chapter (Chapter 4).

Louisa Kennedy was primarily responsible for the remaining parts of the Literature Review, including Place-Based Learning (Section 3.2), Experiential Learning (Section 3.3), Emotions and Cognition Being Intertwined (Section 3.4.1), and Meaning Making as Shared Activities (Section 3.5). Louisa also contributed to revising and expanding the Methods chapter to include the design and procedures for Case Blue.

The Results (Chapter 5), Result Summary (Chapter 6), Discussion (Chapter 7), and Conclusion (Chapter 8) were conceptualised, written, and revised jointly by both authors. Both authors collaborated on data analysis, table and figure design, and final editing of the manuscript, including the Introduction, Literature Review, and Methods chapters.

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1 Introduction

Climate change, biodiversity loss, and social injustice threaten both human well-being and ecological systems worldwide. Global policies such as the Sustainable Development Goals (United Nations, 2015) and the IPBES report (2019) recognise the urgent need for societal transformation to address these challenges. Education, with its wide reach in society, has a powerful role to play in this transformation. Within this context, Environmental and Sustainability Education (ESE) is increasingly positioned as an important tool for supporting more sustainable ways of living and relating to the environment.

Yet despite the increasing integration of sustainability into national curricula, ESE has long focused primarily on knowledge transmission and low-impact personal behaviours, such as turning off lights or recycling (Macintyre et al., 2024). It was assumed that increasing people's environmental knowledge would naturally translate into more pro-environmental behaviour (Kollmuss & Agyeman, 2002). However, decades after the implementation of sustainability policies and environmental education initiatives, the persistence of these crises indicates that knowledge alone does not drive behavioural change. Kollmuss and Agyeman's (2002) "Mind the Gap" framework illustrates this persistent disconnect between environmental knowledge, attitudes, and behaviours. Even when people understand environmental issues and express concern about them, this does not automatically translate into sustainable action.

Emotions have emerged as critical mediators in this gap between knowing and acting. While cognitive and behavioural outcomes have value, Sipos et al. (2008) argue that for ESE to become transformative, it must also engage the psychomotor and affective domains, calling for a holistic "head-heart-hands" (HHH) approach. This means not only teaching about sustainability (head) and practicing sustainable behaviours (hands), but also engaging the emotional dimensions – values, care, and hope (heart) – that motivate deep learning and sustained action. Understanding how emotions shape sustainability learning has become an increasingly important area of research.

The affective domain, where emotions and values are situated, has long been undervalued in ESE research and practice (Grund et al., 2023). However, recent scholarship demonstrates that emotions are not secondary to learning but central mediators between knowledge and action (Chawla, 2020; Immordino-Yang & Damasio, 2007; Manni et al., 2016). Alongside that, young people also increasingly experience eco-anxiety in response to climate change (Hickman, 2024). Ojala (2012) demonstrates that how young people experience and process emotions shapes their pro-environmental behaviour, with hope and trust being important

emotions leading to environmental agency. She distinguishes between denial-based hope, which minimises problems and correlates negatively with pro-environmental behaviour, and constructive hope, which grounds realistic acknowledgement of challenges in the belief that change is possible. Central to constructive hope is trust. Ojala (2016) found that trust in others precedes trust in oneself, meaning that social support and trustworthy role models play crucial roles in cultivating hopeful, action-oriented mindsets among young people. Hickman (2024) further argues that adults have a responsibility to help young people reframe eco-anxiety as a sign of care, validating their emotions while supporting capacity for action.

Research further demonstrates the importance of both self-efficacy and social support in fostering hope. Li and Monroe (2019) found that perceived competence – the belief in one's ability to be effective – predicts hope about climate change among secondary school students. Similarly, Geiger et al. (2024) showed that social support fosters "action hope" which in turn predicts pro-environmental behaviour. Building on these findings, this study examines whether educational programs involving local environmental organisations and passionate volunteers or local environmental professionals may be particularly powerful contexts for cultivating activating emotions. When students work alongside local actors who embody commitment to sustainability through visible environmental action, several conditions align: students encounter trustworthy adults who could become role models of environmental stewardship, observe environmental action in practice, and receive social support that may foster both hope and agency.

Despite growing recognition of the importance of emotions in the field, a significant research gap remains. Most existing studies have focused on quantitative correlations between specific emotions and pro-environmental behaviour among adolescents (Li & Monroe, 2019; Ojala, 2012). Furthermore, no existing research qualitatively examining how local environmental actors shape students' emotional experiences could be found. Despite the growing theoretical attention to emotions within ESE, limited empirical research has explored how affective approaches are experienced by younger students. This research aims to extend existing theories of transformative learning for sustainability that engage the affective domain by studying primary school-aged (6 to 12 years old) students participating in environmental programs that involve local actors.

To better understand how students interpret and give meaning to emotional experiences, this study further draws on Dewey's (1938) philosophy of experience and Pekrun's (2024) Control-Value Theory of emotions. Dewey emphasises that meaning emerges through the ongoing transaction between individuals and their environment, in which experiences are

taken up, transformed, and sometimes reflected upon. Pekrun explains how emotions arise in learning contexts through appraisals of control (Can I influence this outcome?) and value (Does this matter to me?). Together, these frameworks allow examination of how the programs structure learning across the Head–Heart–Hands dimensions, how experiential and contextual conditions shape students' emotional appraisals and how students' meaning making unfolds through their transactions with place, activities, and others.

Given the limited empirical research on younger students' emotional experiences in ESE, this study takes an explorative approach to the broader affective domain, rather than focusing solely on anxiety and hope. This study therefore conducts field observations of two programs in Belgium and the United States.

1.1. Research Aim and Question

This study aims to explore how non-profit environmental programs working with local actors have the potential to facilitate transformative learning for sustainability by engaging the affective domain, that is shared emotional experiences. It therefore seeks to understand how these shared emotional experiences become integrated into students' meaning making in ways that may support this transformation.

To address this aim, the following research questions have been formulated:

Research Question 1: How do shared emotional experiences emerge among primary school students participating in place-based environmental and sustainability education programs facilitated by local environmental organisations?

Research Question 2: How do these emotional experiences shape meaning making among primary school students?

2 Background Information

A case description will give the reader a clear understanding of the observed programs. These descriptions are a result of the researchers' observations, supplemented with information received from the organisation's website, brochures, and conversations with the staff. In this thesis, the two programs are referred to as Case Blue – referring to the river – and Case Green – referring to the conservation work – to foreground them as distinct educational cases in their own right, rather than as representatives of their respective national or cultural contexts. More information about the local context in which these cases are nested, can be found in Appendix 6.

Case Green

Educatief Natuurbeheer (EN) is the educational program of Natuurpunt, a non-profit organisation in Belgium, committed to environmental conservation. Natuurpunt's work includes local biodiversity research, the management of 25,000 hectares of nature reserves, collaboration with (local) governments regarding policy making, and EN, an environmental education initiative (Natuurpunt, n.d.a). The latter focuses on fostering accessible nature experiences for people of all ages, while simultaneously creating a broader societal support for nature conservation (Natuurpunt, n.d.b).

EN is a longitudinal, experiential, place-based learning program designed for upper primary school students (11 to 12 year olds) and is organised in three structured encounters. The program begins with a classroom session introducing students to the changes in their local natural environment and its impact on biodiversity, introducing concepts such as biodiversity loss and habitat segmentation, through games and group work, with a solution-oriented focus emphasising conservation work. Several weeks later, the program continues with a field trip in which students participate in hands-on conservation work within a nearby nature reserve. This work could be clearing cuttings from trees, raking grass cuttings, or removing invasive trees in the forest. This is always real conservation work that needs to be done by the organisation, regardless, and students help them for a day. A third session, later in spring, concludes the program with a return visit heavily focused on sensory activities to enjoy nature. These activities vary per location and educator, but include pond-dipping, meadow flowers investigations, and guided walks with games. The two outdoor sessions are supported by local volunteers from the organisation, or sometimes from other local conservation groups (Natuurpunt, n.d.b).

EN is offered free of charge to students from three provinces thanks to local government financial support. It aims to provide as many children as possible with the opportunity to have positive nature experiences before the age of twelve. Students normally participate once during their primary school career; however, in schools that work with classes bridging several ages, students have the opportunity to join the program several times (Natuurpunt, n.d.b).

Case Blue

This study observed three days of the environmental education programs facilitated by Yampatika, a non-profit organisation based in Steamboat Springs, Colorado, United States. Yampatika's mission is centered around inspiring environmental stewardship through education, which they do through outdoor and place-based learning that comes through experiential learning opportunities for children and adults. The organisation offers different programs that are focused on many things, such as environmental literacy, sustainability, and community engagement. They have things such as school-based programs, outdoor field experience, nature interpretation, summer camps, and service-learning. Each of their programs is designed to connect students with the local ecosystem and the environmental challenges that the Yampa Valley region has and is currently going through (Yampatika, n.d.).

The observations took place in the Yampa Valley region and observed two different groups of primary school students (1st grade and 6th grade). Yampatika educators, all with backgrounds in environmental education and related fields, led the lessons, and they also worked with other local actors. They each took place in multiple local environments (like the Yampa River and Yampa State Park), which allowed students to engage directly with ecosystems in their own community.

During this study, the students worked on ecological investigations and collaborated with their peers and teachers on problem-solving activities. They also did sensory observations, food web and ecology games, nature walks, and had several discussions relating to sustainability and environmental change. During the ‘World Without Fish’ program (Days 1 and 3), the students connected global environmental issues to local fish populations and river systems, and also investigated human impacts on aquatic ecosystems. Then, during Day 2, the students were in three different stations throughout the day: Macroinvertebrates, Beavers, and a Nature Walk.

It was observed that the program consistently had students learn about ecological elements while also emphasising stewardship and their personal relationship with ‘place’. Through hands-on experiences and outdoor education, their approach looks to develop strong

emotional and educational connections for students to the environment in a lasting and meaningful way.

3 Literature Review

This chapter provides a common understanding of the theoretical background used in this study. Transformative Learning for Sustainability (TLS) is foregrounded as a pedagogical philosophy that supports Environmental and Sustainability Education (ESE) to move beyond knowledge transfer and small behaviour changes. TLS is therefore defined first, together with the Head-Heart-Hands framework as a practical way of operationalising this pedagogy. Place-based and experiential learning are then introduced, as it serves as the pedagogical context in which the observed cases are nested. The chapter explores the affective domain of learning for sustainability by zooming in on emotions, followed by a description of how meaning making is understood in this study.

3.1 Transformative Learning for Sustainability

Many policies, like the Sustainable Development Goals (United Nations, 2015) and the IPBES Global Assessment Report on Biodiversity and Ecosystem Services (Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, 2019), call for a deep societal transformation to reverse human impacts on the environment. Education, with its wide reach in society, plays a key role in shaping the next generation's capacity for such a sustainable transformation. Despite the increasing integration of sustainability into curricula, Environmental and Sustainability Education (ESE) often remains focused on knowledge transmission and low-impact behaviour changes (Macintyre et al., 2024). While valuable, this won't be sufficient for a long-term, sustainable future for all. Therefore, a transformation in how humans relate to one another and other-than-humans in a more sustainable way is required (Ball, 1999). A possible way for ESE to foster this shift in worldview is by following a transformative learning (TL) approach.

Mezirow (1997), the founding father of TL, conceptualises TL as transforming "frames of reference" (p. 1) through critical reflection so that they become more inclusive. These frames, built through life experiences, shape how new experiences are interpreted and what is seen as worthy or unworthy. His theory, however, focuses on adult learning, treating childhood primarily as a period in which foundations such as empathy, imagination and emotional control are laid, with true transformation seen as limited by children's capacity for critical reflection.

Illeris (2014) criticises this narrow emphasis on individual, rational cognition and re-orients the target of TL from "frames of reference" towards "identity", integrating cognitive, emotional and social relations. He argues that childhood is dominated by

acquisitional learning and socialisation, with intentional identity transformations more characteristic of youth and adulthood, which helps explain why most TL theories and empirical research are thus focused on adult education.

Nevertheless, research suggests that the seeds of such transformation are already being sown in earlier years. Ball (1999) contests the idea that critical reflection is always needed, showing empirically that emotionally laden, disorienting experiences can shift behaviour and perspective immediately, and introduces the idea of "readiness for transformation" built up through life experiences. Bandura (1997) argues that self-efficacy – the belief in one's capacity to act effectively – is built through mastery experiences in which agency is concretely exercised, suggesting that meaningful action in upper primary ESE is both pedagogically desirable and developmentally formative.

Furthermore, Singleton (2015) argues that children's reflective and self-regulation capacities have been underestimated, and that metacognition can be fostered when enriching experiences are combined with moments for reflection, and that students take up an active, responsible role in their learning. In this view, upper primary can be seen as a transitional period in which emotionally meaningful encounters with environmental issues and others begin to configure how pupils understand themselves in relation to the other-than-human world, laying important groundwork for later, more explicit transformative learning.

This places particular importance on the design of educational experiences. Emotional encounters, opportunities for action, and relational support fostering meaning making emerge as key elements in opening and supporting transformative processes (Ball, 1999; Öhman & Östman, 2007; Rodriguez & Barth, 2020; Glackin & Greer, 2025).

3.1.1 Head-Heart-Hands Framework

The question remains which pedagogical approaches can most effectively weave these elements together in ways that are developmentally appropriate for upper primary students and responsive to sustainability challenges. This is where the Head-Heart-Hands (HHH) framework comes in, first developed by David Orr (1992) and later elaborated by Sipos et al. (2008) to link sustainability education and transformative education through a set of holistic learning objectives embedded in existing curricula. These objectives aim to foster changes in knowledge, skills and attitudes related to sustainability. They integrate aspects such as cognitive engagement, critical and systems thinking (head), practical and action-oriented learning (hands), and affective, value-laden dimensions (heart). When transformative learning is designed by combining these three aspects, it is most effective. While the HHH framework

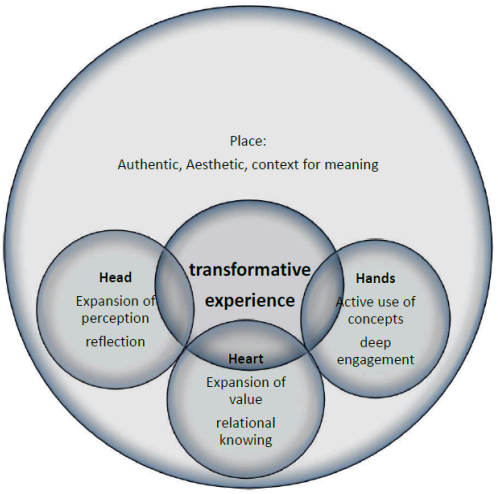
offers a theoretically coherent model for transformative ESE, the affective domain has remained undervalued and under-researched in practice (Grund et al., 2023).

Singleton (2015) expands this affective domain by emphasizing place, especially nature, and its affordance to create authentic, meaningful experiences that foster love, care and engagement regarding ecology, resulting in a shift in perception and values (see figure 1). She highlights the relational dimensions of transformation, aiming for a deeper connection between humans, their community and the natural world, as often practiced in traditional knowledge. This is rooted in a sense of belonging, which strongly influences identity formation and also drives engagement and motivation to transform knowledge into action in learners' communities (Singleton, 2015). Giving more attention to the affective domain will thus affect the other two domains by increasing motivation, including the willingness to critically reflect, and engagement. Altogether, this may support students' environmental agency and the gradual formation of sustainability-oriented identities

She further argues that such transformative dynamics can and should be cultivated already in children through appropriate place-based education, challenging the assumption that transformative, reflective learning is reserved for adults. Her emphasis on love of place aligns with environmental educator David Sobel's ideas, beautifully comprised in his quote "What's important is that children have an opportunity to bond with the natural world, to learn to love it, before being asked to heal its wounds" (Sobel, 1996, p. 9). This suggests that primary-age ESE should first cultivate relational, affective and experiential foundations for later critical engagement. In this thesis, the HHH framework is therefore used as the primary lens to explore how place-based environmental education organised by non-profit, local organisations for upper primary school students brings together cognitive, affective and action-oriented experiences in ways that plant seeds for transformative sustainability learning.

Figure 1

Head, Hand and Heart Model for Transformative Learning Contextualized by Place



Note. From Head, heart and hands model for transformative learning: Place as context for changing sustainability values, by Singleton J., 2015, *Journal of Sustainability Education*. Licensed under Creative Commons Attribution 4.0 International.

3.2 Place-Based Learning

Place-based education provides the context to move beyond predominantly knowledge-based education, as it grounds learning in both the social and ecological environments that students and educators alike inhabit. Meaningful sustainability education should engage learners with the specific ecological, cultural, and social dimensions of place, enabling them to develop a deeper understanding of their relationship to the world around them (Gruenewald, 2003). Place plays its own role in learning, rather than serving merely as a backdrop, for how knowledge is created and experienced. It aims to keep these connections while also encouraging critical reflection on social and environmental issues. Place-based programs prepare students and communities to live and work in ways that have them be active participants in sustaining the ecological and cultural integrity of their inhabited places (Woodhouse & Knapp, 2000). By grounding this specific educational research in the social and ecological dimension of locality, place-based approaches provide an opportunity for experiential learning to unfold, transforming a physical site into a meaningful 'place' through intentional student engagement and interpretation.

One aim of place-based education is to create new narratives and an increased sense of place. Narratives shape how people understand and make sense of the world and its future. By taking time to re-imagine local narratives that connect people with places, new plausible pathways towards a sustainable future are created and agency promoted (O'Brien, 2018; Veland et al., 2018). Narratives of sustainable futures can shift people's perception of nature from a passive recipient of human dominance to an active participant with a reciprocal relationship in society. New narratives and an increased sense of place – the meaning and value people assign to a place – reinforce each other towards more environmental agency (Mannion & Lynch, 2016; Veland et al., 2018).

3.3 Experiential Learning

Drawing on John Dewey's philosophy of experience, Quay and Seaman (2013) discuss how education is fundamentally grounded within experience as an ongoing interaction between individuals and their environments. For Dewey, experience is not a supplemental part of learning, nor is it optional; it is at its very basis (Quay & Seaman, 2013). Experience is not something that happens to a learner, but rather, it is a process of how they make sense of, engage, and interpret situations. Dewey's work challenges assumptions that activity alone leads to learning, and distinguishes between educative and non-educative experiences, the first leading to growth and continuity, and the second having the possibility to hinder further

development (Quay & Seaman, 2013). This distinction is relevant in outdoor and environmental education contexts due to the fact that engaging activities may not necessarily translate directly into meaningful learning. The effectiveness of experiential learning depends less on the activity itself than on how experiences are interpreted and integrated into an ongoing learning process.

The concept of aesthetic experience offers a perspective on understanding what makes certain experiences particularly powerful when this integration occurs. Aesthetic experience refers not only to experiences with art, but more broadly to moments when experience becomes meaningful and emotionally charged (Wickman, 2006). Such experiences do not need to be fully life-changing, but they should touch the individual deeply enough to create a positive personal effect (Ruoppa, 2019). They are characterized by a cohesion of perception, emotion, and understanding that becomes integrated as a whole within a person (Quay & Seaman, 2013), and by immersion rather than passive reception, engaging learners both cognitively and affectively. Emotion, relating to these ideas, is not separate from cognition, but is an organising force in which experience is structured and interpreted (Quay & Seaman, 2013). The following section functions to offer further understanding of what emotions are and how they function within experience and meaning making.

3.4 Emotions

3.4.1 Emotions and Cognition Being Intertwined

Research in affective and social neuroscience supports this centrality of emotions within experience. Immordino-Yang and Damasio (2007) argue that emotion and cognition are an intertwined process, and that emotions play a crucial role in guiding attention, memory, and decision making. Without emotional engagement, students' learning is less likely to be retained or transferred to new contexts outside of the classroom. Emotions work to signal what is important and meaningful to the learner, focusing cognitive resources towards particular aspects of an experience and shaping how those experiences are interpreted. In this sense, emotions can be seen as a basic form of decision making, a back-and-forth between knowing and acting in response to situations (Immordino-Yang & Damasio, 2007). This is in line with the above mentioned idea of Quay & Seaman (2013) regarding aesthetic experiences being the integration of cognition and emotions. Manni et al. (2016) expand upon this within a study of young students, demonstrating how emotions in aesthetic experiences are inherently relational with place, as they emerge through the direct engagement with the outdoor learning environment. For example, students’ emotions such as concern, care, and fascination, were

closely linked to what they were directly learning about, which suggests that the felt emotions and the understanding of the concepts being taught are connected (Manni et al., 2016).

This reinforces the idea that aesthetic experiences, characterized by emotional intensity and engagement, play a key role in facilitating meaningful learning. In the context of ESE, emotional responses may therefore be critical in shaping how students relate and understand environmental issues.

These emotional experiences are not just moments of joy and wonder. A growing body of research also addresses the negative emotional impact of sustainability challenges on young people; however, certain levels of negative emotions could lead to increased agency (Ojala, 2012; Hickman, 2024). Both for the well-being of the student and the planet, there is an urgent need to better understand this wide range of emotions and how ESE can best address them.

3.4.2 Emotions Being Judgements of Control and Value

This research further draws on Nussbaum's (2003) theory of emotions and Pekrun's (2024) Control-Value Theory (CVT). Nussbaum (2003) argues that emotions are inherently cognitive and evaluative: they always relate to an object and carry a value we attribute to it. In the context of ESE, students' emotional responses to environmental degradation signal how important they consider the environment for their own flourishing. Pekrun's (2024) CVT, one of the most empirically robust frameworks for understanding emotions in educational contexts, conceptualises emotions as arising from cognitive appraisals of value and control – the extent to which students perceive a situation as meaningful and feel able to influence or cope with it. Different combinations of high and low values and control generate a wide range of emotions that shape how students engage with sustainability issues.

CVT distinguishes between activating and deactivating emotions, highlighting that emotions differ not only in valence but also in their effects on engagement. When students perceive sufficient agency and personal relevance, positive activating emotions such as enjoyment facilitate deeper engagement. When both control and value are low, negative deactivating emotions like boredom and hopelessness foster disengagement, whereas high value combined with low control can generate negative, activating emotions like anxiety and frustration. This is highly relevant for ESE, where students regularly encounter large-scale challenges that feel important and beyond individual influence.

3.4.3 Negative Emotions in ESE: a Driver or Barrier?

In typical classroom research, negative emotions related to achievement are associated with reduced cognitive engagement and performance (Frenzel & Goetz, 2026). However, emotions related to sustainability challenges, like eco-anxiety, grief at biodiversity loss or moral

indignation at injustice, are emotionally healthy and congruent reactions to a genuine crisis (Hickman, 2024). Depending on their intensity, leaving them unaddressed or invalidated, by contrast, can be harmful (Hickman, 2024; Ojala, 2016). Mild forms of eco-anxiety tend to be activating, whereas severe levels are deactivating and cause distress. Although emotions are inherently internal, the environment shapes control and value appraisals, and therefore emotions. Educators thus need to help students remain in the "activating zone", where difficult emotions can motivate engagement without leading to overwhelm.

The cognitive quality of the learning environment, the emotional tone set by educators, and the degree to which students feel relatedness and (collective) efficacy all influence whether students perceive themselves as in or out of control of meaningful activity (Pekrun, 2024). Crucially, this is a reciprocal relationship: emotions relate to and carry a value for the currently experienced objects or aspects and in turn reshape cognition about the environment. Emotions can then evolve in return due to social context, refocusing and understanding. Hence there is a dynamic reciprocity throughout an experience where cognition and emotion evolve.

In ESE, shifts in perceived control can redirect negative emotions, for example, turning eco-anxiety into constructive hope. Ojala & Bengtsson (2019) show that the coping strategies used in response to climate worry determine whether negative emotions lead to engagement or helplessness, and that these strategies are shaped by the communication patterns of others regarding climate challenges. Supportive, solution-oriented communication predicts problem-focused and meaning-focused coping, while dismissive or doom-laden communication predicts deemphasizing (Ojala & Bengtsson, 2019). This highlights the relational and pedagogical dimensions of emotional support in ESE and calls for a pedagogy that acknowledges difficult emotions and helps students work with them.

When the social environment communicates that others, e.g., scientists, communities, or volunteers, are taking meaningful action, it directly restores perceived control. Li and Monroe (2019) empirically demonstrate that being effective – the belief that society and others have the capacity to act – is the strongest direct predictor of climate hope among students. Ojala (2016) frames this as trust-based hope: young people gain strength to act from believing that others are also taking responsibility. She further demonstrates that this type of hope is positively associated with pro-environmental behaviour (Ojala, 2012). Synthesising this work, Chawla (2020) identifies fostering social trust as a key pedagogical strategy: bringing young people into contact with scientists, activists, and communities who are already working to protect the natural world enables them to see that they are not alone, and that their actions

are part of a greater collective effort. It is this relational embeddedness that sustains hope in the face of environmental crisis (Chawla, 2020; Ojala, 2016).

3.5 Meaning Making as Shared Activities

As defined by Robert Kegan, the pioneer of the concept of meaning making, it is a developmental process through which individuals construct and reconstruct their understanding of experiences, continually redefining their relationship to themselves, others, and the world (Kegan, 1982). This aligns with the aforementioned aim of transformative learning. Öhman and Östman (2007) argue that meaning making is constructed through the interactions between individuals, between groups, and their environments. Rather than meaning being created internally, it can come through the participation in shared activities where individuals have opportunities to draw upon experiences and respond to social interactions. In this process, emotions continue to play a central part, as they shape and reflect value judgments about what is important (Öhman & Östman, 2007). Within this research design, the exploration of the learners' meaning-making process in relation to emotional connection emphasizes the role of interaction with local actors and the inner workings of experience as a whole. As peers and educators interact, their experiences are influenced, and in turn, their meaning-making process.

This constant reciprocal transaction between emotional experiences and meaning making is central in transformative learning, because each episode of making meaning can slightly reconfigure how students see themselves and the world, gradually building a “readiness for transformation” and, over time, opening possibilities for shifts in identity.

4 Methods

4.1 Study Design

This research used field observations to study how shared emotional experiences emerge among primary school students participating in place-based environmental and sustainability education programs facilitated by local environmental organisations.

An interpretative approach fitted this study's aim best to understand the participants' subjective experiences and perspectives, as this approach assumes multiple versions of reality are constructed through context and personal experiences (Braun & Clarke, 2013). The researchers weren't just interested in what emotions arose in the programs, but also in the conditions that evoked these emotions, and what these emotions further elicit. Therefore, field observations were chosen as a data collection method. This allowed the researchers to ground the shared emotional experiences in the context in which they emerge.

Two programs have been researched by two different researchers simultaneously through several observation days for each case. The collected data from the field observations was documented through field notes. This includes informal conversations with involved partners, also documented through field notes. The data was also supplemented with formal interviews with students from one program. The data has been analysed using thematic analysis while iteratively comparing the two cases with the purpose of getting an in-depth understanding of each program, rather than actually comparing the programs with each other. By constantly comparing the two cases, these similarities and differences became clearly visible, enabling the researchers to consider them while analysing and using them as a strength to generate unique, interesting themes. The findings have been discussed alongside existing theories described in the literature review.

4.2 Program Selection Procedure

Two programs have been studied through field observations in different countries: one with Natuurpunt in Belgium (case Green) and one with Yampatika in the United States (case Blue). The researchers worked separately on their case, studying similar pedagogical settings that allow for shared emotional experiences. Both programs have been selected because of the following overlapping criteria:

- non-profit organisations that work with local actors
- place-based education, with the main focus being on local biodiversity
- aligned with the Head-Heart-Hand approach
- primary school-aged students

First, the two researchers each found an organisation in their local area through personal connections and internet searches. After scanning the organisation's websites, one organisation in Belgium and three organisations in the United States were contacted through email. In both countries, one organisation got back to the researchers on time, after which an online meeting was organised with one of the educators of each case individually. Each organisation agreed upon the collaboration for the research.

4.3 Data Collection

Data has primarily been generated through participant observations during field trips and supplemented with formal and informal conversations with staff and students. Eight sessions for field trips were observed during seven days in Case Green and three sessions during three days in Case Blue, for details see Appendix 1. For Case Green observations were complemented by more formal interviews, conducted with students when they were back in school. Parents to the interviewed students gave their written informed consent before the interview and students were asked permission to be interviewed prior to the interview. Students' request of how they wanted to conduct the interview (one-on-one or as a group). These interviews explored how students articulate and reflect upon their experiences and emotions, and how these shaped meaning making. It also allowed for a better understanding of emotions that aren't visibly or verbally expressed and anticipated emotions. The semi-structured interview guide was designed by both researchers before the observations took place and adjusted and translated by the case Green researcher after the first observations to seek answers that were not observed. The English translation of the guide can be found in Appendix 2.

Which available pre-booked visits of school classes at the sites were observed, depended on and were restricted by the agenda of organisations and researchers. Case Green had a wider time window than case Blue, resulting in more observation days, supplemented by student interviews. Ideally, all age groups would have been aligned across both programs. However, because of practical limitations, one group of younger students was included in the observations for case Blue. The following section describes data collection methods specific to each case.

Case Green

The researcher observed eight different class groups participating in the program, of which one was the conservation work day, and the others were sensory activities. While following a smaller number of groups longitudinally would have allowed a deeper focus on individual trajectories, as the program is spread out over a whole school year, and due to the short

timeframe of conducting this research, this was not possible. The researcher therefore deliberately prioritised heterogeneity in school types and settings to better explore how different pedagogical and contextual arrangements shape students' affective experiences.

This includes a special needs school, urban and rural schools and a method school, in different settings, including nature reserves, parks and heritage sites, and in a variety of activities, including conservation work, guided walks and pond-dipping. This allowed the researcher to collect data on a large number of features from the case.

Case Blue

The researcher conducted 3 full-day observations in different areas within Steamboat, Colorado. The first observation took place at an elementary school, where the students went through 4 different activities centred around the Yampatika lesson 'World Without Fish', which students had already participated in previous work with. There were elements of both inside and outside learning, as well as sensory engagement, collaboration with peers, research being conducted in partnership with local actors, and other sustainability topics. The second observation day took place in the Yampa River State Park, where three classes of Year 1 students rotated through three stations, each of which entailed learning and observing parts of the local ecosystem. The final day of observations was with the same class as Day 1, where the class of students went to a fish hatchery and learned from a local actor working there about the hatchery's process, conservation efforts, as well as playing games at the end of the day, connecting to local ecological and sustainability education.

During each day of the observations, as well as before and after, conversations were had with the Yampatika educators, other Yampatika staff, classroom teachers, and informally with the participating students. These conversations gave a better insight into the intentionality of the program, as well as the background of each class and how students engage with the materials and natural settings.

4.3.1 The Role of the Observer

The researchers in both cases adopted the role of a non-participating observer, where they were involved in the activities, but not guiding any of them. Due to the younger age of the participants, they naturally asked some questions to the researchers, and in those moments, the researchers aided if appropriate.

The aim of this study was not primarily to describe the programs themselves, but to examine them through a specific analytical focus or "object" of the case (Thomas, 2021): students' affective experiences during these programs. To study this object, the researchers documented students' shared emotional expressions, interactions, conversations, and physical

actions in situ. The emphasis lay on observable, socially expressed emotions in group contexts. The researchers acknowledge that many emotions are internal and cannot be directly accessed through observation. However, emotions, moods, and attitudes are often expressed through talk, facial expressions, body language, and joint activity, and it was these emotion-laden episodes that were observed and interpreted. Throughout this thesis, the term "emotions" therefore refers to such shared, socially observable emotional experiences rather than to unobservable internal states. It is important to notice that emotions were not interpreted as fixed truths, but rather as contextual, observational indicators. This aligns with this study's interpretivist approach (Braun & Clarke, 2013; Cohen et al., 2018).

During observations, both researchers carried a printed version of the observation guide (Appendix 3), which served as a reference point for focusing attention while remaining open to unexpected events. Each researcher recorded field notes independently in a notebook throughout the observations in addition to the guide. This was used in a non-restrictive way. The researchers sometimes also moved a short distance away from the study and participants to record personal audio that could not be written quickly enough to capture emotional expressions, fleeting interactions, and contextual details. During fieldwork, the two researchers made conscious and consistent efforts to keep in view the broader environmental and social contexts and to maintain a focused attention to specific moments they thought relevant to the study's exploration of students' affective experiences. The researchers thus shifted between a wider observation of the whole group, and sometimes following smaller groups and observing them more closely. The multiple days of observing also allowed the researchers to start with more distant observations to explore all what was happening, then to focus on certain aspects in the following days that emerged as interesting to answer the research questions.

Observations happened in an open, explorative manner where researchers focused on situations related to emotions that naturally occur. This created rich, in-depth data firsthand from participants while simultaneously taking the context into account. By combining observations with informal conversations with students, teachers and educators, the researchers received rich data from three complementing sources, as well as a "reality check" (Cohen et al., 2018, p. 542), attending to differences between observed behaviour and reported words and getting a better understanding of the different aspects of the program that couldn't be observed.

To enhance depth and reflexivity, observations were discussed and elaborated collaboratively between the two researchers regularly. This process allowed for the

development of richer interpretations and reduced the risk of individual and confirmation bias and simultaneously avoided the complexity of consent for video and photographic data in the short timeframe of the research. The duration of observations allowed the researchers to move from descriptive observations to focused and selective observations, and the ability to focus on problems and processes most relevant to the research question increases (Cohen et al., 2018).

4.3.2 Documenting the Data

Both researchers took field notes during the observations, which they then transcribed manually and shared with the other researcher. All interviews are transcribed with the automatic transcribe function from Microsoft 365. They were then replayed to correct mistakes made by the automatic transcriber. The data of case Green was translated from Dutch to English with the automatic translation function of Google Docs. About 10% of the data was checked by this researcher for correctness, because these automatic functions were not always correct. However, it was concluded that, together with the verbal description of each observation, these small mistakes in translation were no barrier to understanding the data for the other observer.

This choice for two cases was guided by interest and practicalities. Each researcher conducted a case study close to their home, where they also taught themselves, which made them very familiar with the local educational context. This familiarity, together with their teaching experience, was seen as a strength to observe the cases in more detail. The strong interest in working with younger students also made it necessary for the researchers to speak the same language as the observed students, as each researcher could conduct their study in their native language.

This double hermeneutic process, where participants interpret their own experiences and researchers then interpret those accounts, is grounded in the study's constructivist epistemology. Both researchers are qualified and experienced school teachers, trained in reading young students' emotions, moods, and attitudes. This experience supported detailed and nuanced observations that went beyond facial expressions, with the researchers taking an active role in interpreting students' experiences. For example, when a student said, "Don't you think we should let that frog go now?", the researchers interpreted this as the student reading the situation as potentially harmful to the animal and expressing care. The subjectivity of the researchers allows for an in-depth understanding of student emotions (both articulated and observed). However, the researchers acknowledge that their positionality could have influenced their interpretations of data slightly.

4.4 Data Analysis

Because the study aims to explore emotions in a specific educational context, and how students interpret and give meaning to them, both the observation descriptives and interviews were analysed using reflexive thematic analysis as described by Braun and Clarke (2013). This method is also supported by Cohen et al. (2018) as a fitting method for analysing natural observations and interviews. The six-phase process of Braun and Clarke (2013) guides the process, which emphasises that this analytical process is not linear. They consist of familiarisation of the data, coding the data, generating and reviewing themes, and producing the report. The interpretative aims within this analysis align with the study's ontological foundation of constructivism, where students are actively giving meaning to their experience, and in turn, the researchers are interpreting this meaning as themes to analyse.

The researchers of both cases started analysing data after the first observations in order to start seeing interesting findings to which they could narrow down focus during the following observations, a method also supported by Cohen et al. (2018). In case Blue, this analysis was done through In Vivo Coding, with codes being created by the researcher as they transcribed the notes, observation guide, and conversations that took place. In case Green, the NVivo software was used to create codes. Complete coding increases validity because it avoids selecting data that fits existing ideas, reducing confirmation bias. Coding in NVivo also gave the opportunity to easily identify relationships between codes. This helped generate interesting themes linking the context, activities, emotions and outcomes. Interviews from case Green were analysed using selective coding to complement the observations. Both researchers followed the same reflexive thematic analysis process but acknowledged that practical differences emerged within the coding procedure due to differences in data volume as well as formatting procedures. Case Blue relied on coding of observation notes and conversations, whereas case Green additionally used NVivo to organise and relate codes across a more diverse dataset, which included interviews.

Formal interviews were only conducted in case Green, making the interpretations regarding students' reflective meaning making more strongly grounded within the Belgian dataset. Case Blue primarily contributes to the observational insights in relation to affective engagement in situ.

After both researchers coded their data, they thoroughly discussed their codes and concluded that there were many overarching themes. Therefore, the decision was made to bring the codes together and generate themes from both sets of data. The findings will be presented in the next chapter, and clearly state which themes were created from which data

set. The heaviness of repeated observations in case Green was taken into account for the analysis by not letting the frequency of these codes automatically translate into centrality. Both cases follow the same overarching methodical framework, as well as the analytical lens, yet the depth and volume of data differ between the two cases due to program structure, access, and timeframe. This study is not aiming for symmetrical comparison, but an analytical resonance across contexts that allows for themes to emerge through a convergence and contrast of the data within the cases.

4.5 Ethics

Both non-profit organisations had agreed to collaborate and expressed interest in the results, creating reciprocity between researchers and organisations, and important ethical considerations (Cohen et al., 2018). The researchers will share findings with the organisations after the thesis is completed.

All school principals and classroom teachers of the observed classes were beforehand informed about the study and the presence of the researcher for observation during their activity. They were thereby able to opt out of the study. None of the schools chose to do this. As data collection primarily involves passive observation, no prior consent from students' families was required. A summary of findings will be shared with participating schools.

No identifiable personal data was collected, nor was any photographic material allowing identification recorded. All participants remained anonymous, and consent was sought from the organisations to use their names in the study. Both organisations are big enough that participating schools or external parties will not be identifiable. Pseudonyms were used for all participants.

Consent was sought from parents to interview the students, and students were reminded about their right not to answer or stop the interview at any time. A group of 2 and 3 students asked to do the interviews in groups with their peers, which was accepted by the researcher. All interviews were conducted in the students' school, contributing to their comfort.

Power and positionality inevitably influence research (Cohen et al., 2018). With this methodological framework, this impact was considered minimal. However, the researcher's presence during the observations could possibly have affected students' behaviours minimally, which was considered inevitable and with little impact on the results or students well-being. As the interviews sometimes involved negative emotions, the researcher responded with empathy, leaning on their past teaching experience to handle these situations.

This study acknowledges limitations in capturing full longitudinal meaning-making processes and transformation due to its short timeframe. It therefore aimed to identify patterns

in how emotions function within meaning making in specific contexts, rather than reconstructing complete processes.

4.6 Quality and Validity

This study does not aim for statistical generalisation. Instead, it works to provide rich, context-sensitive insights into how emotions are experienced within specific educational settings. This approach responds to gaps identified in the literature, where previous research may rely on elements that do not capture the complexity of emotional experience or rely on different educational contexts. In this exploratory, two-case design, validity is therefore not understood in terms of measures or representativeness, but in terms of the credibility and trustworthiness of the interpretations developed from the data (Thomas, 2021).

However, as this is a study with two cases where data collection and part of the analysis are done by two different researchers, reliability should be discussed. The researchers have taken measurements to align their methods of data collection and analysis. The cases were selected because of similar program designs, the data collection was supported by an observation guide, and both researchers started with a complete analysis of their data to make sure they had a deep understanding of their findings. On top of that, both researchers regularly discussed their observations with each other, and the second researcher interviewed the main researcher to enhance the quality and reduce subjectivity in the observations. Nevertheless, there are limitations to this parallel study approach regarding the subjectivity of the researchers. By acknowledging them, the researchers hope to give the reader a clear understanding of the study, including these limitations.

Another way to contribute to the quality of the study is the triangulation of data. By combining observations of multiple days of the project with different groups, together with interviews and informal conversations, the credibility of the findings is strengthened.

Because of the uniqueness of each case, the aim is not to generalise the findings. However, each case can contribute to a growing body of research, and thereby it does contribute to a broader generalisation of theory (Cohen et al., 2018).

5 Results

5.1 Answering Research Question 1: Shared Emotional Experiences

Across the different activities and programs, a wide variety of shared emotional experiences were observed. Two themes were identified that directly speak to the emerging emotional experiences. See Table 1 for an overview including the identified sub-themes.

Table 1

Themes and Sub-themes Overview Relating to the Emergence of Shared Emotional Experiences

Themes	**Sub-themes**
Activities that Spark Specific Emotional Responses	
	Encounters with Living Beings and Water
	Games, Competition, and Playful Tasks
	Physical and Embodied Tasks
	Stories, Facts, and Talking-Based Segments
Human Support that Tunes and Redirects Those Emotions	

When zooming out and linking these emotions to specific activities, clear overarching patterns emerged. An overview is given in Table 2. First, we describe the activities that spark specific emotional responses, followed by the human support that tunes and redirects them.

Table 2

Overview Activities and Emotions

	Type of Activities			
	Encounters with Organisms	Games and Playful Tasks	Physical and Embodied Tasks	Knowledge transmission
Examples of Activities	Pond-dipping, nature walks, magnifying glasses, meadow investigation	Bingo, fish crime game, invasive species game, capture the flag	Clearing invasive trees, investigating, and observing living organisms	Stories, Facts, and Environmental Narratives about i.a. local degradation, restoration successes, and fish recovery

Affect	Awe, empathy, pride, care, curiosity	Excitement, motivation, joy, competition Sometimes: distraction, chaos, worry	Pride, achievement, social cohesion and collaboration, discomfort, tiredness, boredom	Concern, hope, surprise, worry Sometimes: boredom, disengagement
Outcome	Empathic concern for non-human beings	Engagement and collaborative meaning making	Self-efficacy and agency	Future-oriented thinking

Adult support that tunes emotions

Modelling wonder, knowledge, and care

Redirecting chaos to calm

Acknowledging discomfort without amplifying fear

5.1.1 Activities That Spark Specific Emotional Responses

Both programs offer a wide variety of activities, and it became apparent that different activities provoked different emotional responses. These responses in turn acted as entry points into environmental learning and meaning making, or sometimes, as a barrier. Patterns across both cases will be described.

Encounters With Living Beings and Water

In both cases, activities that offered the possibility for encounters with living organisms like plants and animals triggered strong emotional responses. Both programs explicitly designed hands-on, sensory activities to facilitate these encounters. The educators were thereby often directing students' attention to observe animals and plants in detail, to use other senses than their eyes to observe the surroundings, or by slowing down activities and energy for encounters to be felt more deeply. This resulted in strong emotions of awe, astonishment, and enjoyment, as well as curiosity and arousal, as seen in the following examples:

> *Case Green, Day 5, field note*: Then Ella (educator) hands out the small insect jars and instructs them to catch insects in them. The students catch large spiders. Sam tells them that it is probably a male because it has "punching bags" on its front legs. The students are very engaged and ask if that has to do with the size of the spider. Sam points out more

details to the students: "Have you seen the hair on its legs yet?" The students look more closely and look at each other
with a face full of wonder.

Case Blue, Day 2, field note: Walking to station 2 at the edge of the river, the educator has all of the students stop before passing through the gate. As they approach, the students are all talking loudly, yelling, running, kicking rocks, etc., and the educator turns to them, asks them to pause, and asks them about their 5 senses. Then they have the chance to share what they were thinking about. The educator may be using this to ground them in the place they are in and to calm them before entering the area, as not to disturb the flora & fauna, and for potential risk hazards next to the large flowing river.

Encounters with animals also evoked sensitivity and admiration. This was often paired with signs of worry and care, indicating empathy, where students asked questions about the well-being of the animals or directed their peers to handle animals with care.

Case Blue, Day 2, field note: At the river's side during the macroinvertebrate station, students are in small groups interacting with a bucket of water that comes from the river they are next to. They are exploring the different species they see and comparing them to an identification chart provided. As the students are working, many of them are expressing care for the living things in the water. Example: "Be careful! If you pick up the rock too fast, the bugs will get scared" (Case Blue, student)

Case Green, Day 5, field note: Suddenly, the student says, "Look, a frog!" Another student: "Oh, another one!" Both students immediately try to catch those frogs and hold them. (...) The students are having a conversation about the frog with each other: "What would frogs eat?" (...). Frogs keep appearing, so the students keep catching frogs. At a certain point, a student has been holding a frog for quite some time, and some students say, "Hey, where is the freedom for the frog?" "You are hurting him!" They are quite concerned about the frogs by now.

Catching animals during the pond-dipping activities or spotting traces of beavers in the landscape also triggered emotions of pride.

Case Green, day 4, field note: After the activity, the students continue searching for animals. And when they meet their peers who are in the other group again, they are super enthusiastic to show each other their finds. The other group also starts talking very enthusiastically about the beavers and the beaver tracks they found.

Case Blue, day 3, field note: While at the first tanks of fish, students observed at the fish hatchery, they are all going back and forth through the tanks and asking questions to the

guides, but also to their friends and the general teacher. They are sharing their excitement with their peers, and asking curious questions like “How many are in the tank?” “How long will they be here?” “Are there more fish that we get to see?”

Encounters with plants were seen to evoke playfulness and creativity.

Case Green, day 5, field note: Further along the walk, they stop, and Elaine (volunteer) explains that nature is full of solutions. She refers to cleavers and explains that Velcro comes from it. She explains how cleavers work. From then on, the students started playing a lot with the cleavers. They are also a bit sneaky and secretly stick cleavers back onto each other. Then they all start sticking them onto the teacher's backpack. They clearly have a lot of fun with that.

This was also seen when students made little packages from leaves filled with flowers for their teacher. Students used an acorn head or a blade of grass as a whistle or fooled each other by letting peers smell stinky plants.

Case Green, day 6, field note: And then Thomas (educator) shows the students how to make a whistle from a rowan twig. He incorporates mathematics and pulls out his bag of tricks to cut the bark and slide it off the twig. The students are hanging on his every word. There is a lot of amazement when a student is actually able to blow the whistle.

Moments of excitement were sometimes paired with moments of chaos. While pond-dipping, students were heavily focused on the sensory activity of catching animals with the net, and less focused on the task of identifying the animals. These moments often went from chaos at the start to more structure and rest throughout the activity, when the first strong emotions settled, and attention was redirected by the staff.

Case Blue, day 1, field note: The students are in small groups conducting their ‘fish research’, they were able to pick what group they would be in (observed that many went with their friends in the class) and during the end of their allotted time (as shown on a timer on the classroom wall), many of the groups were already completed with the assignment and talking about other things or doing other tasks around the classroom. The educator has noticed this and is going to each group, asking them more investigative questions, and pushing them to further their research.

The opposite was observed during the ‘living walls’ activity in case Green, where a unique species of lizards lives. There, students' attention went from focus to distraction during an open task.

Case Green, day 3, field note: The students head out; the wall is several hundred meters long, so they can spread out and seem very fascinated by the lizards. After a while,

however, they become quite distracted and start mostly roughhousing. The students had about 20 minutes to explore without an extra assignment. The atmosphere however remains playful and warm.

During guided walks, there were several moments where students were distracted from the given task. However, looking closer at what the students were doing, most of the time they were still engaged with nature: discussing with a peer, looking through the binoculars, or playing with cleavers. This was observed similarly in both cases:

Case Green, day 4, field note: The students were also given a pair of binoculars, and they are very engaged with them, spotting things in the fields. Sometimes, while Ella (educator) is giving their explanations, it may seem as if some students are distracted, but those students are still busy with things related to the guided walk, such as looking through the binoculars.

Case Blue, day 2, field note: During the station where students are on a nature walk in the state park, they are each given binoculars. They are taught how to use them and practice with them before their walk. When they begin the activity, the majority of the students are very excited and running to different things to observe, possibly missing what the educator is saying and teaching about different species, yet they are still very engaged. At the end of the walk, the educator goes over each thing that was mentioned during the walk and asks questions to the students- a high level of engagement during the activity.

These patterns indicate that close encounters with local species not only generate momentary emotions but also function as early experiences of empathic concern and care-oriented responsibility towards non-human beings in students' own environments.

Games, Competition, and Playful Tasks

Both programs implemented a wide variety of games and playful tasks throughout the observation days. These could be divided into three categories: structured content games, team-based competition, and open-ended play in and with nature.

Across both cases, structured content games were used as a didactical method for knowledge transmission.

Case Blue, day 3, field note: After the hatchery visit, activities with the local actors concluded, the students gathered again with the Yampatika educator after lunch. They play the first game- each student is given a laminated card with a species (flora or fauna) of the local area, and then they pass a ball of yarn around to show the lines of energy transfer between each species. Example: a fish eating a plant (pass the yarn), then a fish eating that same fish (pass the yarn), and so on… at the end, there is a full observable 'food web' as

created by the yarn passing. The educator then talks about the fish species they have been learning about, as well as the ecological aspects of energy transfer.

These games pull students in through excitement and challenge. Students were visibly animated and eager to participate when ecological content was presented in game form.

Often, games involve competition: groups collect points, compete in teams, race to complete a task, or present findings in front of others and compare performances. This could be seen within both of the cases:

Case Green, Day 1, field note: The students can place an animal in the aquarium if the species isn't already there. This is a strong motivator to use the identification chart. In the beginning, this is quite difficult, so the volunteer passes by each group to get them started with the identification chart and goes through the steps. This creates a bit of competition and motivates them to actually use the chart, because the students naturally want to contribute something to the aquarium that isn't there yet.

Case Blue, Day 3, field note: One of the closing games of the field trip to the fish hatchery- students are in a field, and they each given a card with the fish species they are (either invasive or noninvasive to the area), they have to race across the field to collect 'food' which is colored plastic balls (the color of the ball is specific to each species), this is a timed activity. The majority of the students are highly engaged and racing each other to beat the others to the 'food sources'- similar to the scientific material they have been learning about ecological efforts from species regarding food loss.

These activities typically evoke high energy, laughter, elevated arousal, motivation, and group pride, but can also momentarily narrow focus to winning rather than understanding. This is an example from the observations:

Case Green, Day 1, field note: For the sensory walk, the students get a bingo card with local species of fauna and flora. Two students are enthusiastic when they spot a plant from the card. When Peter (volunteer) then directs the whole group to this plant, the students are upset because they feel like they lost an advantage in the game.

Overall, these playful activities shaped the emotional climate. Students laughed together, tested social boundaries, and appeared increasingly relaxed in the outdoor settings. This low-stakes joy appeared to make it easier for them to later engage in more demanding ecological tasks and created warm, relaxed group dynamics.

Physical and Embodied Tasks

These activities were particularly apparent in case Green, where students helped Natuurpunt with conservation work. This required physical labour, sometimes in

uncomfortable circumstances. The emotional responses were divided. Some students showed high motivation and enthusiasm, clearly comfortable in the natural environment. Others showed discomfort and boredom. Growing self-efficacy and enjoyment through practice were also observed:

> *Case Green, Day 6, field note*: After a demonstration from Thomas (educator) on how to eliminate invasive species by digging them out or using a bark spud, the students start doing the same in small groups. (observer starts following a group of students who, during the introduction, said that they "hate the forest" and that "the forest is only good to let the dog out") The students start digging out a medium-sized tree. After they successfully do that, they want to move on to a big tree. Their teacher passes by and questions whether they should try to dig that tree out and mentions they could use the bark spud instead. The students reply: "No, I'm sure we can do this!" and enthusiastically start digging out the tree. They're jumping on the shovel, pulling, and pushing. They are laughing a lot and keep working for a long time, even though it becomes already quite early apparent that they won't be able to dig that tree out.

During an informal conversation at the end of the day (Case Green, day 6, continued field note) with the student who expressed negative emotions towards the forest at the beginning of the day, she mentioned a clear shift in value:

> *Case Green, Day 6, informal conversation*: That was the girl who said at the beginning of the day that she doesn't like the forest. And now they were talking about it when asked about the day, saying that it was really different from what they had expected and that nature and the work Natuurpunt does is very important. Together with her friends, she discussed how important it was to take care of nature and help it.

Such reflections suggest the emergence of stewardship identities, as students came to view physically demanding conservation tasks as part of what it means to contribute to protecting local nature.

Stories, Facts, and Talking-Based Segments

Both programs were grounded in place-based environmental concerns. Hearing these stories provoked feelings of concern, interest, and surprise when they were concrete and local. These responses point to future-oriented environmental concern that is grounded in specific local places rather than abstract global issues.

> *Case Blue, Day 1, field note*: The students are working together as a whole group to do a timeline activity with the Yampatika educator. They are each given a laminated paper with a year and an event on it, specifically regarding an event about the river in the area. After

putting themself in order and sharing what their event is (the educator followed each event with more information about the details), they were asked what this meant to them personally, and what they thought it meant for larger environmental issues outside of their community. Afterwards, the educator related this to the importance of students' future careers being within environmental fields due to changing issues.

Pride and joy were observed when students heard about restoration successes. This was observed during a speech from the chairman of the local department of Natuurpunt in Case Green, when a ten-year partnership with the school was celebrated by agreeing on a lifelong sponsorship:

Case Green, Day 7, field note: Philippe (...) addresses the group with a speech and says that thanks to their help, the nature reserve has grown. He speaks of an additional 200 ares that they have been able to purchase thanks to good maintenance. He emphasizes that the students should be proud. The students applaud, and many students are smiling. After the speech, the school signs a contract to seal the sponsorship. The teachers read it aloud. “The first part: we are going to commit to protecting nature.” All students spontaneously shout: “Yes!”

And in case Blue, as students were being called upon to consider future careers in environmental sectors and learning about their direct impacts to parts of ecology, similar emotions emerged.

Also, in both cases, fun facts about animals and plants were a common thread throughout all the activities, resulting in moments of wonder, surprise, joy, and intrigue.

On the other hand, these talking-based segments risk losing effective engagement if not anchored in concrete examples. For example:

Case Blue, Day 1, field note: Students are working in their small groups to conduct ‘fish research’, as the educator is talking about different specifics that could affect the fish populations as well as the different local organizations they will be using to research these effects, some of the students groups seemed to lose focus- possibly because they were no longer engaging with the material directly?

During longer explanations, or when stories involved abstract processes that students appeared not to fully understand, their attention sometimes drifted away.

Overall, these emotions were observed across all students and ages. During informal conversations with the classroom teachers, differences in background were explained, where some students showed an already existing emotional relationship with nature, and other

students encountered natural elements they had never seen before. In general, this did not seem to impact the types of emotions that arose, according to the teacher.

5.1.2 Human Support That Tunes and Redirects Those Emotions

In both locations, adults were often observed modelling emotions of wonder and enjoyment, mostly by verbally expressing them. In case Blue, as the class walked to the riverside, the Yampatika educator asked the students to pause and reflect on how they were a part of the natural setting. Saying that:

> "Just like you have your indoor classroom every day, where you are now is my classroom. But an outdoor classroom has some different parts. Before we go to the river, close your eyes and tell me what your five senses are feeling." (Case Blue, Non-profit Organisation Educator, Field note)

Asking questions to invoke their senses, which was met with encouragement and engagement from the students, appeared to set a tone of heightened focus before continuing with the activity.

> *Case Blue, Day 2, field note*: After the students participated in the questions and activities before entering the station by the riverside (the educator asking them to think and share about their 5 senses), the students were no longer separated from the group or talking as loudly (the educator specifically commented on loud human noises making animals around them scared). They entered the station in a much calmer and focused manner.

In case Green, it was observed that a classroom teacher felt negative emotions relating to animals during the pond-dipping lesson. When the students caught the first invertebrates, the classroom teacher said to the observer: "Oh, when I see all those little creatures, I just feel urgh, I'm already getting stressed out, huh" (Case Green, Day 2, informal conversation notes). Similar expressions were observed throughout the activity that indicated a feeling of discomfort from the teacher.

However, other teachers were observed being very involved with the students' enthusiasm, supporting students who showed negative emotions towards nature. There were several moments where the teachers approached these moments by allowing these emotions without giving them too much attention. Across observations, such adult responses appeared important in either amplifying or softening students' emotional reactions, and in helping them to re-engage with activities after moments of discomfort or distraction. Overall, these findings suggest that both program designs evoke strong positive activating emotions leading to high focus, engagement, and interest.

5.2 Answering Research Question 2: Emotional Experiences Shape Meaning Making

Across all data, three overarching themes capture the relation between emotions and meaning making. Each theme has three sub-themes (Table 3). Each theme will be explained with empirical examples from both cases. Similarities and differences across cases will be discussed for each theme.

Table 3

Themes and Sub-themes Overview for Shared Emotional Experiences Related to Meaning Making

Themes	Sub-themes
Experiential Learning for Deep Environmental Understanding	
	Investigating with bodies and tools
	Moving, slowing down, and using the senses
	Acting out ecological relationships
Social Learning Strengthens Engagement and Meaning Making	
	Educators modelling environmental care
	Local actors as knowledge holders
	Collaborative learning among peers
Place-Based Relationships Foster Meaning, Connection, and Care	
	Local Place as the Foundation for Learning
	Sense of Place Through Sensory Immersion
	Reinterpreting Familiar Places Through Ecological Knowledge

5.2.1 Theme 1: Experiential Learning for Deep Environmental Understanding

While themes that speak to research question 1 described how different activities evoked emotional responses and how adults tuned these in the moment, this theme comprises how hands-on, sensory, and embodied participation supported students' ecological understanding and meaning making. Across both cases, students appeared to make sense of sustainability concepts most strongly when they investigated, moved, and talked through ideas in direct interaction with local environments and others.

Investigating with Bodies and Tools

Both cases strongly emphasize inquiry-based learning, where students were provided with tools or were invited to use their own bodies to investigate natural elements.

> *Case Green, Day 7, field note*: After that, there is a tree scavenger hunt in pairs. One student with special needs has a personal assistant with him. Earlier, it became apparent that it was hard for this student to engage with the activities and work with others. For this activity, he and his assistant go a little further into the forest. The assistant is blindfolded; the student leads him to a tree. The student is very enthusiastic and laughs a lot during this activity. Other students are also engaged and are laughing a lot.

During these active inquiries and hands-on activities, students appeared highly engaged and motivated. They frequently showed their findings to educators and peers and expressed interest in continuing the tasks. Educators then linked these observations to ecological and sustainable concepts such as biodiversity, invasive species, or river health, and students posed questions that connected their observations to broader sustainability issues, often emerging from feelings of empathy, care, and concern regarding the living world. For example:

> *Case Blue, Day 3, field note*: The students are going through each of the fish tanks and asking questions to the guide, educators and peers. The students are constantly moving back and forth through the area and tanks, and asking what the next site they will visit will be. They are also answering questions from the hatchery guide and educator that are linked to previous lessons on the fish populations, as well as the local river system.

Where both cases strongly link the place-based inquiry to local sustainability challenges, case Blue also links this to broader, global sustainability challenges. In case Green, the outdoor sessions strongly emphasized sensory activities through which students learnt about local fauna and flora, sometimes missing an explicit link back to earlier discussions of biodiversity loss and habitat fragmentation from the classroom-based sessions. At the same time, the emphasis on sensory exploration – increasing aesthetic experiences – supported the development of local ecological literacy, as students learnt to recognise species, habitats, and ecological relationships in their own surroundings, and a deep connection with the living world.

A clear difference was observed in the way the programs shape these inquiry-based learning activities. In case Green, students frequently handled plants and animals directly through pond-dipping, conservation work, and playful interactions with natural materials. This is similar to the classes for the younger students, in case Blue; however, a shift was noticed in their grade six activities. There, hands-on work took the form of using tools, props,

and digital devices, such as laminated timeline cards, skit scripts and costumes, and online 'fish fact' files on laptops. Ecological interaction was therefore partly mediated through research tasks, role-play, and local actors, with fewer moments of students physically touching plants and animals during the observed lessons.

A further difference was that, in case Green, conservation activities such as removing invasive trees formed part of ongoing management work that needed to be carried out in the nature reserve. Whereas in case Blue, the grade six hands-on components (e.g., research tasks, games, and hatchery visit) were primarily designed as a way for students to learn about a collective understanding of local ecological challenges through the area's history.

> *Case Blue, Day 1, field note*: During the 'fish species' research activity, followed by the 'fish crime' activity, the students are tasked with using what they have learned from the educator, and then using their laptops to do further research. The information they are recording on their 'fish findings' sheet is later connected to their local river system and other local actors that work on the environmental issues they are learning about.

However, across both programs, students appeared to build ecological understanding through active investigation and discussion, even when the 'hands-on' dimension was structured differently.

Moving, Slowing Down, and Using the Senses

In both cases, educators deliberately designed moments where students were asked to slow down and engage their senses to become more aware of their surroundings. During nature walks and outdoor investigations, students were encouraged to close their eyes to listen, to notice smells and textures, and to observe visual details in the landscape. In case Blue, students were guided through 'five senses' activities before starting investigations, while in case Green, educators regularly paused walks to invite students to quietly look and listen in particular spots.

The connection between these practices and students' environmental understanding became apparent during a pond-dipping activity in case Green:

> *Case Green, Day 2, field note*: One small group is working on a small pond. Three of them are working together smoothly and are very concentrated. They scooped into the pond for the first time and emptied their nets into the container. They take a quick look, and then one student goes back to the pond to continue scooping. Ella [educator] arrives and looks at the tray and says: "You already have a lot of little creatures!". The students still standing at the tray react surprised. Ella: "Yes, lots of little ones!" (pointing at micro-invertebrates in the tray). The students are looking more closely at the tray now and start seeing all the tiny

little creatures. Student 1: (direct attention to the other student who is already back at the pond) “Hey, apparently we already have millions of little creatures!” The student runs back to the tray and looks at the tray intently and discovers all the little creatures. A lot of surprise is observed.

This moment suggests that directing attention to detail helped students notice more in their surroundings, which in turn stimulated more inquiry. Building on the emotional engagement described in Theme 1, these sensory practices appeared to support students in describing organisms in greater detail, e.g., by naming specific features, habitats, or behaviours, thus providing them with more ecological language when explaining concepts such as biodiversity or habitat quality

These sensory activities did not automatically lead to explicit discussions of sustainability. On several occasions, students commented on how ‘nice’ or ‘fun’ a place or activity felt and moved on without it being observed that they connect these impressions to ecological processes or environmental issues. This suggests that while slowing down and using the senses can deepen attentiveness to place, their contribution to environmental understanding possibly depended on how educators followed up these experiences in subsequent dialogue or students’ prior nature connectedness. For example:

Case Blue, Day 2, field note: After the students came to the end of their nature walk station, a student was speaking about how they liked the nature walk and did not connect it to what they had learned, but more so that they liked being outside and how they wanted to come back. A student said: *“Yeah, I liked the bird nests we saw. But I want to take my mom back here to see these yellow flowers.”* (Case Blue, informal conversation, student).

Acting out Ecological Relations

During the skit, in case Blue, students were given different characters and roles to act out, e.g., an animal, a news reporter, a local rafting guide, an ecological scientist, and got a script and props to support them.

Case Blue, Day 1, field note: After the students worked on the timeline activity, they were given roles (students were able to choose based on what they were interested in), and a script as well as props the then practice their lines and put on a short ‘play’ about the issues within the local area regarding fish populations and ecological challenges within the river systems.

Through this performance, students created connections to different ecological relations.

During the first session in case Green, groups played a game where they could help clean nature to protect biodiversity. Although this specific activity was not observed by the

researcher, the students directly reflected on it months later during the interviews conducted in this study, clearly remembering the activity and the sustainability challenges that it revealed.

Overall, these embodied, sensory activities appeared to increase students' engagement and interest, which in turn seemed to strengthen their motivation to learn more about and deepen their understanding of the sustainability challenges and possible responses.

5.2.2 Theme 2: Social Learning Strengthened Engagement and Meaning Making

One of the core differences between the two cases becomes apparent in this theme. In both cases, educators working for the program guide the students through the activities. In case Green, educators have a background in biology and nature conservation, whereas in case Blue, they have a background in ESE. Additionally, both cases further shift knowledge transmission from 'teacher - student' to 'wider community - student' by working with external but local stakeholders. In case Green, the program works closely with local volunteers who are actively engaged with the organisation, while in case Blue, the local actors of the fish hatchery are environmental professionals. These qualities appeared to shape students' experience and meaning making in different ways, which are described in sub-themes below, followed by a section on collaborative learning between peers.

Educators Modelling Environmental Care

Across both cases, the professional and experiential backgrounds of the program educators shaped how environmental work was modelled to students.

Case Green

The program educators of case Green did not have formal teacher training, but both worked for the organisation for several years. This experience appeared to translate into clear instructions, activities that were well adjusted to the age group, and a strong sense of how to read the group's energy. The positive relationship between educators and students became visible on several occasions:

> *Case Green, Day 1, field note*: The students arrive at the nature reserve by bike. (...) When they arrive, Ella [educator] is waiting to greet them. The students recognize her and greet her back. A few students come over to have a chat with her, and one student eagerly shows Ella their new glasses.

The educators had qualifications in biology or conservation work, and this strong content knowledge became apparent through the strong emphasis on species identification and local fauna and flora, as seen in this example:

> *Case Green, Day 7, field note*: Ella [educator] brings the students back together for the next activity. And as they stand in their small group, a tiny moth caterpillar suddenly dangles

down by a silken thread. Ella says, "Oh, look, we have a visitor." Everyone looks at the little caterpillar with great admiration. Ella explains why it is coming down — probably danger up in the tree, maybe a small spider — so it can lower itself down. Then it can climb back up again, and the students watch it, and one student said, "Don't touch it!" Completely in the moment, it says a lot about Ella's knowledge that she knows all that. And the students find it incredibly interesting to watch, filled with wonder.

Students routinely turned to the educators to check identifications and ask a wide range of questions, to which the educators could usually provide an answer. They modelled enthusiasm and care for our environment both verbally and in their actions. Students were seen copying this behaviour, e.g., when carefully returning pond water so as not to harm any remaining organisms or eggs, or by carefully walking through the meadows not to step on flowers or frogs. The educators also acknowledged limits to their own knowledge and provided the students with tools to access information by showing a mobile app for identifying fauna and flora. Through these interactions, environmental work was implicitly modelled as practical, place-based stewardship, combining specialised knowledge of local species with everyday practices of care in students' own landscapes.

Case Blue

The Yamapatika educators demonstrated strong backgrounds in EE, ESE, and outdoor education. This became clearly visible through their professional experience, and personal enthusiasm for the material and working with students were present and also appeared to directly affect student engagement.

Case Blue, Day 1, field note: Before the students began their activities for the day, the Yampatika educator introduced themselves and talked about their background specific to environmental work and their love, knowledge, and work for the local area.

Rather than outside instructors delivering information and facts, they were active environmental practitioners who showed their care and passion for the local environment, species, the Yampa River, and sustainability issues within the Steamboat area and beyond.

The observed enthusiasm and passion were specifically seen in the educators' consistent use of ecological stories, personal connections to the local environment, and place-based examples that the students could also connect to.

Case Blue, Day 2, informal conversation with educator: They spoke about their previous experience in schools throughout the United States and how they began to notice the large difference in their overall understanding or general care for the environment. This was one of the things that motivated her to go into this specific line of work. Noting that they feel

like the work they are doing allows more students to have the opportunities to not only go outside, but to learn about things specific to where they live.

Within the lessons, when scientific knowledge was shared, it was frequently connected to the real environmental issues affecting the students' own communities and realities. During the World Without Fish program, the Yampatika educator and fish hatchery staff discussed the impacts of climate change, invasive species, habitat degradation, and river temperature changes in ways that directly connected to the Yampa River and students' everyday experiences. Throughout these talks, the students became more engaged, especially when they were involved either with living species (such as fish or birds) or with changes to a local area that the students personally knew.

Taken together, the case Blue educators tended to model environmental work as a professional, knowledge-rich practice explicitly addressing local sustainability issues, while the Green educators mainly modelled environmental care as knowledgeable, hands-on stewardship of local species and habitats. In both cases, students experience educators as trusted, caring adults whose expertise and enthusiasm shaped how they related to the living world. The intentionality of using pedagogical approaches such as hands-on ecological investigation, inquiry-based questioning, and experiential learning highlighted the educators' strong experience working in these programs. The activities were not students sitting and listening to information and facts; they were centered around students being encouraged to investigate, observe, and ask questions to their peers and teachers so that understanding could be built in multiple ways. Taken together, the educators' professional backgrounds shaped not only the structure of the activities but also students' collaborative and individual meaning making throughout the program.

Local Actors as Knowledge Holders

Case Blue

Students' meaning making was also supported by other local environmental actors. During the students' research of fish species and solving a 'fish crime', they explored information directly from the websites of the Colorado Parks and Wildlife, U.S. Fish and Wildlife, and the U.S. Forest Service organisations, and later worked with the hatchery staff. Students not only learnt scientific content from these local actors but also appeared to develop trust in their confidence in expertise about the ecology in which they were studying. For example:

Case Blue, Day 3, field note: During the students tour of the fish hatchery, they are not only asking questions constantly, but they are excitedly going to the local actor (leading hatchery staff) and ask questions such as "How are there so many fish in one tank?",

"What do they eat?", "Where do they go after this?", "What do you feed them?", "Why are they swimming in only one direction?".

Such questions were met with care and enthusiasm. The socially supportive learning environment that was built through the collaboration of the educators, the students, and these local actors continued to construct further and deeper understanding of the ESE elements within the lessons.

Case Green

Volunteers in Case Green played a role similar to the program educators, with many showing extensive local knowledge and modelling enthusiasm and care for the living world in playful ways. Students showed enthusiasm when they recognised returning volunteers, indicating an evolving relationship, as seen here:

Case Green, Day 5, field note: Ella [educator] has the whole group standing around her in a meadow. Ella says: "Look, there is Samuel [volunteer]. You remember him, right, from the work day?" The students are enthusiastic, smiling and weaving: "Yes, Samuel!"

Students' comments and interactions suggested that they experienced these volunteers as people who embodied environmental stewardship beyond personal gain. This became clear in a student interview (case Green). One student described feeling "so bad" when seeing videos of environmental destruction and talked about "bad people" destroying other species. When the researcher then asked them if being in the program changed something about that feeling, the same student responded "(...) that gives me a really good feeling, because there are (...) of course, people [volunteers] who want to help nature, and that is nice to be able to be a part of that" (student, interview, 11 years old). They exemplified a social norm where environmental care has high priority. When asked how they felt about working with the volunteers during the interviews, other students expressed admiration and inspiration for their work. They described it as "really brave that they want to do all of this for free and free their time for nature" and appreciated that "it was nice working with them because they helped us and we could ask them anything". Several students said they now also want to help keep nature clean themselves. One student commented that it was good to learn from the volunteers, because "they know much about nature and my teachers know much about mathematics", indicating that they distinguished between different kinds of expertise in their learning environment. These moments suggest that working alongside volunteers helped this student reframe their sense of humanity and that they themselves could be part of that caring group.

Working with volunteers also has limitations. During one guided walk, a volunteer appeared insecure when responding to a student's concern regarding a seemingly polluted

river flowing into the sea and replied: “Yeah, our sea is anyway polluted too”. At this moment, a potential opportunity to connect the place-based inquiry to sustainability challenges was not taken up. Overall, volunteers in Case Green model a strong nature appreciation and local stewardship, together with a more hesitant engagement with complex, systemic sustainability issues.

Whereas case Blue students were encouraged to access information from formal environmental institutions and develop trust in these professional knowledge holders, case Green students primarily encountered volunteers as local experts on species and places to whom they could relate, offering students an example of ongoing citizen stewardship that they could potentially join themselves, with more varied engagement in discussions about larger-scale sustainability issues.

Collaborative Learning Among Peers

Across both programs, strong collaboration between peers and between peers and staff was observed throughout all activities. Within the different spaces where the lessons occurred, the positive and collaborative elements when collectively learning about and exploring ecological experiences were consistent. Throughout the lessons, the students were continually sharing observations, pointing out species, and talking about prior experiences relating to what they were doing.

Collaborative learning worked to strengthen place connection by offering many opportunities for students to socially construct environmental understanding together.

5.2.3 Theme 3: Place-Based Relationships Foster Meaning, Connection, and Care

Local Place as the Foundation for Learning

Across both programs, students consistently connected the environmental learning to places they knew and had interacted with in the past – whether through recreation or family experiences, reflecting an explicit place-based program design. In case Blue, activities took place in areas such as the Yampa State Park, the Yampa River, and the Colorado River, while in case Green, each group visited a nearby nature reserve, always close enough to the school for classes to arrive by bike or foot. Surrounding ecological systems were therefore not presented as distant or abstract concepts.

This familiarity surfaced as a way to connect emotional engagement described in Theme 1 with students’ meaning making throughout the programs. When students already knew the place they were learning about, they often appeared eager to interact with the information and activity. At the same time, in case Green, some students did not know the nature reserves beforehand. Through the place-based activities, these students were given the opportunity to

become familiar with new local areas, which created possibilities for them to develop greater knowledge of their own living environment and to return to the areas for recreation in the future. For example:

> *Case Blue, Day 1, field note*: During the timeline activity the students are frequently linking previous experiences to the local areas and were excited to share their previous knowledge about a species being discussed. The students commented they had "been there before" or shared memories relating to hiking, rafting, or spending time near these waterways.

Through this heightened engagement, they were able to make further connections with the timeline facts to actual lived experiences. Similar moments happened during the nature walk and the hatchery visit observations, where the students were consistently connecting new information to their previous experiences with the local environment. These repeated references in the activities suggest that the knowledge being shared was not about an unfamiliar ecosystem, but rather they were interpreting familiar places through new ecological information.

In case Green, the program is explicitly designed for students to become more familiar with the local area through repeated visits: first, during the conservation work day, and then again during a later sensory activity. This allowed students to see and reflect on the results of their work. It appeared to increase their emotional connection with the place and the importance of their work.

> *Case Green, Day 4, field note*: During the walk, they also pass the meadow where they worked on the conservation work day. Ella [educator] stops there and they look at the work they have done: clearing the branches from the pruned willows and then they had to stack all that wood. The students were surprised to see the changes in the meadow: already so many new twigs were growing on the stacks of wood and the meadow was covered in flowers.

A student from this group expressed seeing the difference in an interview with enthusiasm and wonder: "That was a whole field full of branches, so we had to clear it all away, and now there were tons of flowers!" *(*Student, 11 years old, interview, Case Green). Pairing this return visit with a sensory activity where they caught and observed insects appeared to strengthen their emotional connection to the place and their sense that their earlier efforts had mattered for nature. In this way, the work and its visible results were linked to the sustainability concept of biodiversity, as students encountered a greater variety of plants and insects in the area they had helped manage.

When asking another student from that same class what it means to be able to help nature during the conservation work day in case Green, they answered: “I actually think it is quite important, because, well, a lot of animals live there, and the bigger nature is, the better for the world” (Student, 9 years old, interview). Here, the student articulated a simple but powerful form of relational value, connecting the extent of “nature” in their local area to both animal well-being and a broader sense of what is “better for the world”.

Because this work formed part of the organisation’s ongoing management of the reserve, students were not only learning about biodiversity but also participating in the actual care of a place they could revisit. This kind of before-and-after experience appeared to nurture a sense of place in which students linked visible ecological change to their own prior actions, reinforcing the idea that local landscapes are dynamic and responsive to human care.

Furthering this connection, the local relevance of the environmental issues shared with students appeared to increase the emotional investment. There were particularly strong reactions when students learnt about ecological degradation or threats to species and waterways in their own community. For example, in case Blue, during discussions about river closures due to rising temperatures, students displayed visible concern and attentiveness because these closures directly affected their recreational experiences and local environment, which was observed when a student said: “I knew they closed the river because of the fish! That’s why I don’t go as much with my family in the summer now (Student, 11 years old, Case Blue, field note).”

This was observed through student conversations where they talked about not enjoying when the river was closed, students sharing their displeasure with the environmental issues causing this, and audible reactions such as sighing and an upset disposition when talking further about this issue. Environmental problems became meaningful not only because they were important ecological issues, but also because they directly impacted elements that students personally valued.

Sense of Place Through Sensory Immersion

Sensory immersion also appeared to shape students’ relationship with place by making them more aware of their own impact on non-human life. In both cases, educators reminded students to move quietly, avoid picking flowers or hitting trees, and follow ‘leave no trace’ principles. For example:

> *Case Blue, Day 2, field note*: During the nature walk station, the educator and general classroom teacher were continually (calm mannerisms) were redirecting students to stay on the path and think about their impact on the environment, even though they were so

excited. Saying phrases like “I know that tree look really cool from here, but we don’t know what we will stop on off the path on your way there, so use your binoculars to look at it from here, please’, “I hear you guys (students) saying how much you love the animals we are seeing! Make sure we are remembering that this is their home, and if we are loud, it will scare them away”.

Students responded by adjusting their behaviour and reminding each other of these practices. These moments suggest that sensory engagement supported students in seeing themselves as members of the local ecology, not just observers visiting for the day. Reciprocally, an increasing sense of place was also noted to provoke emotions of excitement and curiosity.

Reinterpreting Familiar Places Through Ecological Knowledge

In case Blue, the presence of intentional place-based methods within the lessons was evident, as well as working to deepen and transform how students understood those places. These familiar environments became connected to larger environmental systems and sustainability concerns as students learnt through ecological investigation, storytelling, and conversations with local actors. The program not only exposed students to local places, but they also transformed how students understood them.

Learning about elements such as invasive species altering ecological balance (both cases), hatcheries contributing to not only recreation but conservation (case Blue), and how water temperature affects trout populations (case Blue), grew students’ understanding of how the nature reserves are not only a recreational space they are connected to, but also a complex system that is shaped by human and non-human interactions. For example:

Case Blue, Day 1, field note: During the timeline activity and the skit, the students were consistently referring to information they were learning to things they had previous knowledge on because it affected their personal life (e.g not being able to swim in the local river because they close it for periods of time to protect the fish populations). They were also making these connections while showing interest in learning about new ways the areas they know were being negatively affected by different ongoing things in the area.

This reinterpretation of a place students had previously expressed their connection to was especially seen when listening to their connections of new ecological information to future environmental concerns. In both cases, students asked questions that extended from personal concern, e.g., river closures affecting recreation, to broader ecological futures, e.g., how human actions affect fish populations or how beaver dams shape nearby ecosystems. This shows that students link previous and new knowledge and think critically. For example:

Case Green, Day 6, field note: Then Thomas [educator] begins his explanation, detailing that they are going to work in the forest and uproot the American cherry because it is non-native. He explains why that non-native tree ended up in Belgium: “(...) but actually, it is a real pest, because it leaves no room for the other trees, so they call it the forest plague”. He mentions that, unfortunately, the experiment failed, as is often the case when people think they can alter nature. After the explanation, the students enthusiastically start removing the invasive species. Most students are very engaged, working hard, sweating and getting dirty.

This repeated use of local examples built upon students’ abilities to connect more abstract sustainability concepts to more concrete realities that involved their growing knowledge. As they made connections between concepts like habitat degradation to local species, climate change to rivers and landscapes, and ecological interdependence in general, students' environmental learning appeared to be emotionally meaningful and personally relevant.

5.3 Result Summary

This study explored how shared emotional experiences emerged among primary school students participating in place-based environmental and sustainability education programs and how these experiences contributed to meaning making. Across both cases, emotional experiences and meaning making were closely intertwined, with emotions acting as entry points into learning, reflection, and engagement with environmental topics. The following section summarizes the relationship between the research questions, and the results themes.

Both programs actively shape emotions through their pedagogical design, consisting of direct encounters with living beings and water, playful activities, embodied tasks, stories and fun facts. The combination of these activities brings a wide variety of shared emotions and moods, from joy and engagement to awe and curiosity, to pride and hope. Negative shared emotions such as concern were observed when environmental degradation stories were linked to local places, and distraction or boredom arose when talking-based pedagogies were used or when one activity lasted for a longer period of time. Social support and the possibility for hands-on, embodied activities contributing to the restoration of the environment contributed to a shift from negative to positive emotions. These findings are described in section 5.1 (Answering Research Question 1: Shared Emotional Experiences) through the activity-based patterns in section 5.1.1 and the role of adult emotional support in section 5.1.2, address research question 1.

Addressing research question 2 (5.2 Answering Research Question 2: Emotional Experiences Shape Meaning Making), three themes capture how these emotional experiences

shape students' meaning making. Through experiential learning tasks (section 5.2.1), high levels of engagement were observed, resulting in agency in the present moment through a willingness to learn and act in the activities. Social learning through educators, local actors, and peers (section 5.2.2) further strengthened meaning making, with conversations and reflection revealing an emerging future-oriented agency, where students showed increased willingness to continue caring for their environment. Place-based relationships (section 5.2.3) deepened students' emotional connection and sense of environmental responsibility, particularly when they could observe the tangible results of their own actions. Across both programs, these three dimensions together support emerging forms of stewardship and agency. In case Green, this was translated to practical conservation work in the local area, while case Blue shaped this more towards the possibility of careers in environmental care and seeing students as future decision-makers.

Overall, the findings suggest that emotional experiences functioned as an important mechanism through which students engaged with environmental learning. Emotions were not separate from meaning making but were deeply embedded within experiential, social, and place-based learning processes. Across both cases, emotional engagement supported the development of environmental understanding, strengthened relationships with local places and communities, and contributed to students' emerging sense of care, responsibility, and agency toward environmental sustainability.

6 Discussion

The findings demonstrate patterns in how different pedagogical approaches evoke different affective reactions in students. Awe and empathy arose during organism encounters, curiosity and enthusiasm through playful activities, pride during conservation work, and concern when hearing local environmental stories. These emotions were shaped by the affordance of the place, which created repeated opportunities for aesthetic and sensory experiences, and by social interactions among participants and modeling pro-environmental behaviour by adults. This mix of different emotionally laden experiences appeared to support a deep understanding of the ecological content and meaning making, important processes in Environmental and Sustainability Education (ESE). When interpreted in light of transformative learning theories (Illeris, 2014; Sipos et al., 2008; Singleton, 2015), these findings suggest that such meaning making can reorient how students relate to the world, indicating emerging transformation in their ways of seeing and valuing human-environmental relations, with environmental agency as one possible expression of this shift.

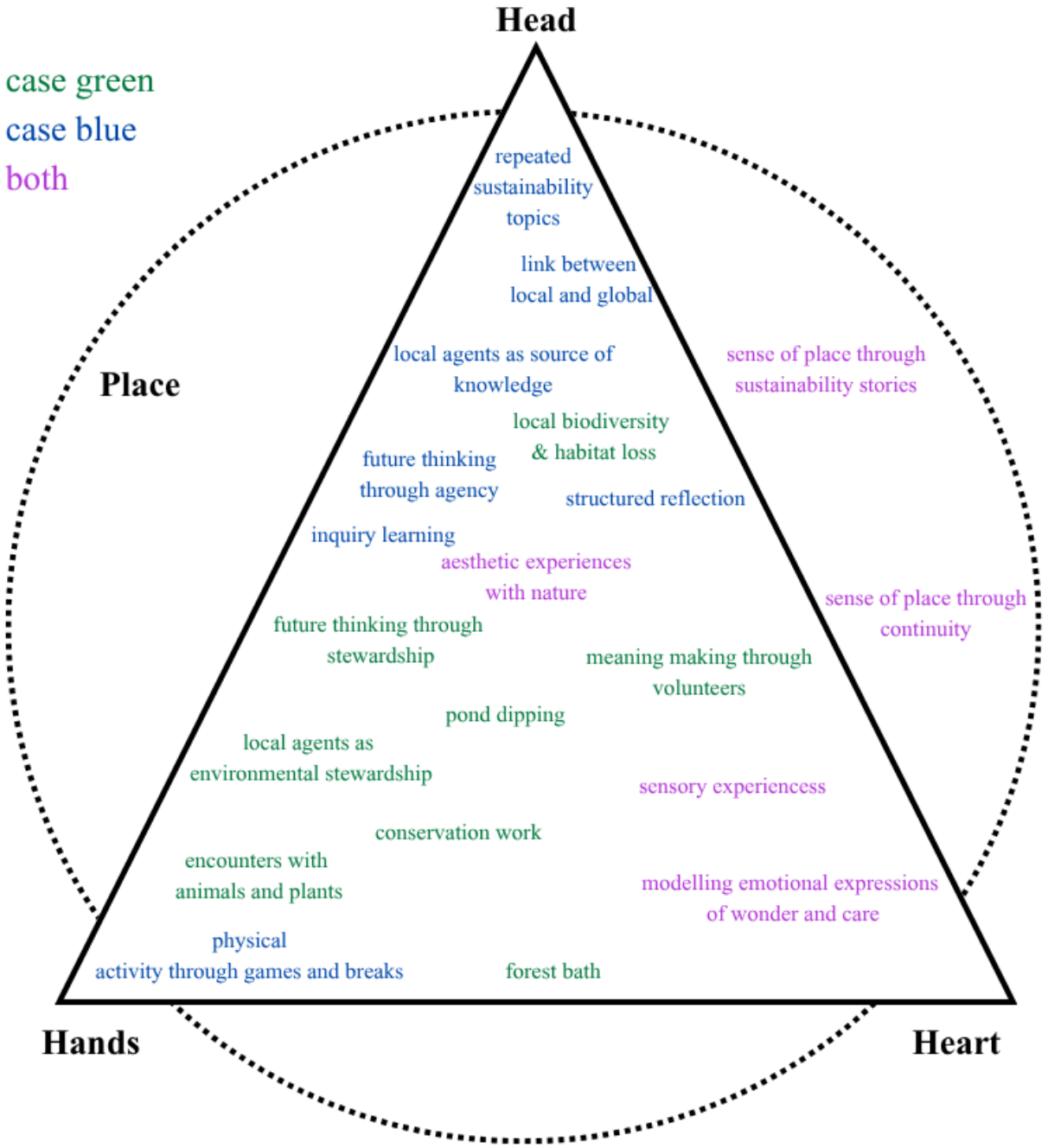
6.1 Head-Heart-Hands as an Organising Lens

The Head-Heart-Hands (HHH) framework conceptualizes transformative ESE as the integration of cognitive, practical, and affective dimensions in mutually reinforcing ways (Sipos et al., 2008; Singleton, 2015). Because emotions do not operate in isolation – they shape and are shaped by cognitive understanding and practical action – the discussion examines all three dimensions to capture the full picture of how affective engagement drives meaning making and possibly fosters transformation.

Both programs clearly include all three aspects of this framework, embedded in place-based education and explicitly oriented towards sustainability. By putting the two programs next to each other, distinct emphases within the HHH framework became apparent. While both programs strongly engage the affective domain, case Blue highlights the cognitive domain through explicit sustainability concepts and theoretical inquiry, whereas case Green foregrounds the practical domain through conservation work and repeated physical encounters with local species. The main areas of divergence are the degree of integration between ecological knowledge, affective engagement, and practical action, and the depth to which local environmental issues are connected to broader systems. How these emphases shape meaning making and agency will be discussed in the following paragraphs. An overview of the main similarities and differences is given in the graph below (Figure 2), based on a framework designed by Sipos et al. (2008).

Figure 2

Pedagogical Landscape of both programs based on the organising principle of head, hands, and heart



Note: Adjusted from Sipos, Y., Battisti, B., & Grimm, K. (2008). Achieving transformative sustainability learning: Engaging head, hands and heart. *International Journal of Sustainability in Higher Education*, 9(1), 68–86. Original figure name: "A "pedagogical landscape" of sustainability and transformative learning models based on the organizing principle of head, hands and heart" (p. 77)

6.2 Looking at Emotions Through a Control-Value Lens

The Control-Value Theory (CVT) (Pekrun, 2024) offers a productive lens for interpreting the emotional patterns observed across both programs. CVT conceptualizes emotions as arising from learners' appraisals of how valuable an object or situation is to them and how much control they feel over its outcome. Across both programs, many activities appeared to invite appraisals of high value and high control: hands-on investigations, games, and practical conservation work centred on local species and places that mattered to students, while providing clear tools and structures for acting competently. In these situations, students frequently displayed enthusiasm and enjoyment.

Curiosity was also frequently observed. Within the CVT, curiosity arises as an epistemic emotion when students perceive a knowledge gap as both worth closing (value) and closeable (control). The living organisms and novel ecological phenomena in both programs appear to have created precisely these conditions. There were indicators that care deepened over time, suggesting a gradual increase in the perceived value of the natural environment and of participating in its care. During the conservation work day, perceived control appeared to increase as students mastered the tasks, which was reflected in growing expressions of pride and confidence.

Negative emotions, such as worry, arose when students heard stories about environmental degradation in their local environment, indicating a high level of value and a low level of control. This high level of value is aligned with Nussbaum's (2003) theory, according to which emotions are linked to an object and tell us something about the importance of that object for one's own flourishing. Boredom arose during longer explanations, consistent with Pekrun's (2024) proposition that boredom results from low perceived value, regardless of control level.

However, in some instances, distraction by competing sensory stimuli appeared to precede conscious appraisal, suggesting that in outdoor ESE settings, the immediate environment may capture attention before value or control judgements are formed, a dynamic that CVT, designed primarily for formal classroom contexts, does not specifically foreground. More about external influences on control and value, and emotions that motivate action, such as hope and pride, will be discussed in the identity and local actors' section.

Overall, the findings support Nussbaum's and Pekrun's shared premise that emotions are shaped by how individuals appraise their situation and suggest that program designers can intentionally cultivate positive and activating emotions by structuring activities that offer both high perceived value and achievable levels of control.

6.3 Place-Based Learning, Aesthetic and Embodied Experiences

6.3.1 Emotion and Knowledge are Integrated Through Aesthetic Experiences

Aesthetic experience, as understood in this study, is not limited to appreciating nature as beautiful (Wickman, 2006). Students describing organisms as “interesting”, “gross”, “important”, or “cool” were doing more than expressing feelings; they were organising their attention and assigning personal meaning to environmental experiences. The findings suggest that aesthetic, embodied experiences play a central role in how students engage with ESE, highlighting that learning becomes meaningful when it is connected to students’ lived experiences as well as ecological contexts (Woodhouse & Knapp, 2000). This supports Deweyan (Dewey, 1934) ideas on aesthetic experience, namely that learning emerges through the active interaction between individuals and their environments. Sensory encounters consistently appeared to create immediate emotional responses (e.g., excitement, discomfort, curiosity), which students then drew on to shape their interpretation of the activities. In practical tasks and encounters with living organisms, students were frequently moving between emotional expression and scientific observation simultaneously. Manni (2016) and Quay and Seaman (2013) describe this as the integration of emotion and knowledge within environmental learning, where emotions are closely tied to the specific contents and situations students engage with. Aesthetic experiences can thus be seen as the mechanism through which feeling and knowing come together into meaning making.

6.3.2 Local Places for Aesthetic Experiences

For each case, the role of local place was especially important to both the aesthetic and embodied experiences of the students. Both programs grounded the learning for students in their local ecosystems, reducing the distance between them and large environmental problems. Gruenewald (2003) argues that place-based approaches are a way to encourage learners to reflect and create relationships between education and the places they inhabit. The findings within this study strongly depict this process, showing how by connecting activities to students’ communities, lived experiences, and familiar outdoor spaces, they connect deeply with the ecological content.

The findings of this study also support Sobel’s (1996) argument that younger learners benefit from developing emotional bonds with nature before confronting large-scale environmental crises. As found in both cases, emotional engagement was most visible when the students were interacting directly with living organisms, local habitats, and restoration activities in their communities. Rather than focusing on distant global issues or abstract environmental statistics, students were given several opportunities to learn and create a

meaningful relationship with nature. Emotional attachment and care appeared to emerge from these relationships and bond to a specific place. Moreover, students were introduced to environmental problems alongside examples of restoration and stewardship, something that is key in preventing disengagement or hopelessness, according to Ojala (2016).

The findings suggest that aesthetic and embodied experiences in which emotions are embedded serve as a bridge between the three dimensions of sustainability learning. Students' emotional reactions, such as awe and care (heart), often sparked curiosity and inquiry (head), while practical engagement allowed students to turn those emotions into participation and action (hands). The embodied experiences supported both ecological understanding as well as care and confidence within students, increasing a sense of place and a possible greater willingness to engage with environmental issues in the future. This aligns with the pragmatist perspective that emphasises that learning and agency deepen when students participate in meaningful situations that are not detached from their lives and are based on knowledge acquisition alone (Quay, 2013).

6.4 Social Learning Supports Meaning Making

Social interaction and meaning making are tightly interwoven in both programs in this study, where the former is one of the mechanisms through which the latter occurs. The transformative learning theories all point towards the importance of social interactions to create deep learning, possibly leading to transformation (Illeris, 2014; Sipos et al., 2008; Singleton, 2015). Sipos et al. (2008) emphasise the importance of community collaboration and democratic learning, through which students' participation and responsibility for their own learning are strengthened. In both programs, students learnt from and with local actors, and all activities involved collaboration between students, resulting in sustained peer discussion.

Illeris (2014) shifts the focus of transformative learning from frames of reference to identity, explicitly integrating social relationships into how learning is understood. Most of the students in this study were in the transition period between childhood socialisation and the formation of identity during youth. The programs offer opportunities for socialisation through the involvement of the different local actors who embody environmental stewardship. It thereby possibly contributes to what Ball (1999) calls "readiness for transformation," as well as for emerging identity formation through new, sometimes disturbing experiences, and supported reflection. We could thus conclude that regardless of the students' development and readiness, the programs have the potential to foster the (trans)formation of students' identities. Singleton highlights how this critical reflection is possible for younger students through

collaborative learning with each other and place, as these encounters are “mirrors” for our own reflection (Singleton, 2015). She thereby emphasises how loving a place increases the motivation for this type of self-reflection. Illeris (2014) argues that such motivation should not be offered by the educators but found by the students themselves. This is where the strength of the programs in this study lies: they actively shape moments in which students can discover motivation through aesthetic experiences, place-based activities, and reflection through interaction with others. This was observed through a high level of engagement, even in uncomfortable situations, conversations expressing care and future-oriented emotions, and reframing their perceptions. This lays the groundwork for a new narrative: one where environmental stewardship is the social norm, encompassing hope, agency, and a high nature value (Veland et al., 2018).

This social meaning making was further supported by the longitudinal design of both programs. Multiple encounters actively shape continuity, an important aspect of educational experiences according to Dewey (1934). This was observed through students enthusiastically recognising returning staff and giving meaning to their hard work when seeing the results of the conservation day. These moments increase the opportunity for students to link new knowledge to previous experiences through the transaction between students and the environment, an important aspect of meaning making that shapes future experiences (Öhman & Östman, 2007). Emotions play an important part in this, as positive emotions like joy increase participation and motivation, which increases reflection (Singleton, 2015). Here, the affordances of the place are again an important factor. It functions as an affective anchor: the element that binds cognitive, social, and emotional dimensions of the experience into a coherent whole (Singleton, 2015; Gruenewald, 2003). Moreover, where different academics have different ideas on whether transformation happens through critical reflection (Mezirow, 1997; Illeris, 2014; Singleton, 2015) or all of the sudden (Ball, 1999), the findings of this study reveal that meaningful meaning making also transforms identities. Students shifting from “hating the forest” to actively engaged to contribute to the forests’ health because of social support and place-based, hands-on activities demonstrate this.

6.5 Different Pathways Toward Transformative Learning for Sustainability

The findings support previous research demonstrating that participation in material and ecological practices is central to developing engagement and environmental responsibility (Caiman & Lundegård, 2014). Both cases demonstrated students' increasing levels of confidence and willingness to contribute to environmental activities, which can be interpreted not only as agency but as part of a deeper reconfiguration of identity and responsibility

(Illeris, 2014). However, the forms of this reconfiguration differ in relation to scale, educational emphasis, and participation.

Within case Green, students' agency emerged through direct stewardship and participation in local environmental restoration efforts. Students were physically engaged in conservation work alongside local volunteers. By successfully removing invasive trees in the forest, environmental action became more tangible and personally meaningful, and students developed feelings of usefulness and responsibility. These findings align with Bandura's conception of self-efficacy, where mastery experiences in meaningful tasks strengthen beliefs about one's capability to produce desired outcomes (Bandura, 1997), and with Li and Monroe's (2019) argument that perceived competence and societal efficacy are key psychological factors fostering climate-related hope. Through these moments, students begin to see themselves differently, as people who can and do care for local nature – which Illeris (2014) would describe as a shift in identity orientation. In this sense, the conservation activities in case Green can be seen as supporting a transformative shift in how they relate to environmental issues.

However, this form of stewardship-based agency has limitations. Gruenewald (2003) notes that place-based approaches, when focused primarily on local ecological stewardship, risk remaining insulated from the broader social and political dimensions of environmental issues. Without explicit connection to systemic causes, local care can fall short of the critical ecological literacy he envisions as part of a fuller place-based pedagogy.

In contrast, case Blue emphasised future-oriented environmental understanding and inquiry in relation to ecological systems thinking. Throughout the activities, students were consistently encouraged to investigate ecological relationships and connect their local observations and knowledge to broader environmental processes, as well as contexts outside their community. Agency within this context appeared more closely connected to ecological understanding and future environmental roles rather than immediate, local restoration participation. Students were frequently positioned as informed decision-makers – imagining themselves as scientists and environmental stewards. This approach aligns more closely with Gruenewald's (2003) vision of a critical pedagogy of place, in which local ecological experience serves as a springboard for understanding the broader social and political systems that shape environmental conditions. However, in contrast to Case Green, this approach then provides less practical tools for students to continue with pro-environmental behaviour beyond the program, as future careers are still far away for students.

However, the findings of this study also suggest that cognitive engagement did not consistently generate strong feelings of agency. Moments in which environmental problems were framed at a large scale or in abstract terms occasionally produced feelings of emotional distance and uncertainty. Ojala (2016) argues that knowledge about environmental crises can contribute to anxiety and disengagement when learners lack visible pathways for action. When broader environmental challenges were raised without immediate opportunities for participation or concrete pathways for action, the tension among students was evident.

Together, these findings suggest that neither purely practical nor purely cognitive approaches to agency are sufficient on their own, and that the affective domain plays a mediating role in making both meaningful.

6.6 Local Actors Elicit Trust

Across both cases, agency appeared the strongest when social support, emotional engagement, and participation all converged. Activities that combined student emotional significance with achievable forms of participation often created feelings of curiosity, enjoyment, hope, and pride. Pekrun's (2024) Control-Value Theory framework helps to explain this process as students appeared to be more engaged when they were doing activities that were both meaningful and perceived as achievable. This is in line with Bandura (1997), who regards confidence as a quality that is partly developed by observing others successfully engage in meaningful action. The educators, local actors, and volunteers all played an important role in both cases, as they were modelling environmental agency in a realistic and socially valued way.

As one student, in case Green reflected, feeling that "there are people who want to help nature, and it is nice to be part of that" illustrates precisely how contact with volunteers shifted the student's self-positioning to a participant in a caring community, which is the relational mechanism Ojala (2016) identifies as the foundation of constructive hope. The findings demonstrate that environmental agency develops through multiple pathways involving emotional engagement, social support, and student participation. In case Green, a stewardship-oriented agency was grounded in local care, while case Blue was focused on inquiry-based and future-oriented forms of environmental agency.

This distinction is also visible in the trust students expressed in the local actors. The foundation of that trust, in case Green was the visible commitment of volunteers – people freely giving their time and skills to protect the local environment. In case Blue, this trust was rooted in the expertise of governmental and local institutions that constantly work on finding solutions to protect the environment. It can thus be concluded that both program designs

evoke trust within students, case Green on the micro-level and case Blue on the macro-level. When students saw volunteers or professionals as trustworthy, committed adults, the human-nature narrative shifts towards a more caring one, which echoes Chawla's (1998) finding that role models are central to significant life experiences shaping environmental sensitivity. Li and Monroe (2019) argue that perceived societal efficacy is the strongest predictor of climate hope. Ojala (2016) argues that constructive hope – grounded in trust in others already taking action – is a precondition for students' own sense of agency – the idea that seeing others act restores perceived control and makes individual action feel meaningful rather than futile. This links the observed levels of trust and agency in this study. Together, the findings suggest that environmental agency should be understood as multidimensional rather than a singular educational outcome.

It should also be noted that the levels of trust generated from this collaboration are dependent on the local actors' capacity to acknowledge and constructively redirect students' negative emotions, and there have been small moments in case Green where missed opportunities regarding this were observed. It can thus be concluded that working with volunteers does increase trust, but it also brings challenges, as supporting students' negative emotional responses is a demanding task that requires pedagogical skill not every volunteer has been trained for. This study thereby contributes to the growing body of research on the importance of emotions in ESE and provides program organisations with a concrete basis for developing volunteer training around emotional pedagogical support.

6.7 Non-Profits in Relation to Affective ESE

Whereas ESE is inherently critical of political and social structures, the role of formal education is to "maintain the present social order" (Stevenson, 2007, p. 1). In addition, formal education typically separates cognition from emotions and divides its learning content into discrete subjects. Non-profit organisations are structurally positioned to do what formal schooling often struggles to do, in that they embed sustainability learning in authentic local contexts with real ecological stakes and community relationships. The findings of this research suggest that these are key conditions necessary for genuine affective engagement.

Non-profit organisations and non-governmental actors play an important role in navigating the contemporary landscapes of ESE, as they operate outside of the typical institutional boundaries of formal schooling. One major strength is their structural flexibility, where they are not bound to rigid, hour-by-hour timetables or standardized testing routines; they are able to pivot their designs to accommodate real-world environmental phenomena.

Another strength of operating outside these formal schooling boundaries lies in the programs' capacity to de-emphasise the dominance of scientific knowledge over aesthetic experiences, a concern central to Dewey's (1934) argument that aesthetic experience is itself a form of knowing, inseparable from feeling. In *Experience and Education*, Dewey (1938) further critiques formal schooling's over-reliance on abstract, transmissive knowledge and argues for continuity between lived experience and learning. By stepping out of the classroom, away from schoolbooks and tests, and by foregrounding local actors as knowledgeable experts, these programs successfully counterbalance that knowledge hierarchy. However, given the short duration of these programs within students' broader school careers, their cumulative effect remains limited. Furthermore, it was observed that some classroom teachers, even within these programs, tended to frame scientific and measurable outcomes as the primary learning gain for the students, which further illustrates the difficulty of fostering transformative learning within an existing system.

Another benefit of the non-profit programs is that their operational agility is tied to their local embeddedness and community connection. Because each of the non-profits is rooted within the social fabric of their regions, they hold deep localised expertise in relation to community-specific ecological vulnerabilities. Rather than students studying abstract global issues from a textbook, they are engaging in experiential opportunities that are personally relevant and visible in a more immediate way. Hands-on encounters with local ecological realities transform passive knowledge transmission into active, sensory-rich meaning making, directly fostering a profound relational connection to place (Mannion & Lynch, 2016).

6.8 Seeds for Transformation

Across observations, students occasionally demonstrated deep learning and meaningful meaning making, evolving in emerging forms of environmental stewardship and agency. This can be interpreted as early transformations in how they position themselves within environmental stories, connecting ecological knowledge to questions of care, responsibility, and human impact (Illeris, 2014; Singleton, 2015). These were expressed through students' emotional responses, collaborative discussions, high motivation and engagement, and sometimes reflections about both their personal future and the future of local ecosystems. In both programs, students came to see themselves less as detached observers of big environmental challenges and more as participants within environmental systems, suggesting a shift in their ways of relating to the world rather than only focussing on low-impact behavioural intentions.

Future-oriented thinking further illustrates how these programs functioned as ‘seeds’ of transformation. In case Blue, students were observed linking current ecological issues to possible environmental careers and expressed interest in continuing to learn how to care for nature, while in case Green, observations, informal conversations and interviews revealed that students were positioning themselves in relation to environmental futures even when structured reflection was limited. From a transformative learning perspective, these shifts in imagined futures and identity orientations can be seen as beginning transformations rather than fully consolidated ones (Illeris, 2014; Ball, 1999). Given the short duration of the programs and their positioning within formal schooling, the findings therefore suggest that such non-profit-led ESE initiatives are best understood as planting seeds of transformation: they create emotionally charged, socially supported, place-based experiences that can reconfigure students’ meaning making and open up new possibilities for how they relate to environmental issues in their lives.

7. Further Research

Because of the explorative manner of this study, many directions regarding future research have been identified, particularly in relation to the conditions in which emotions occur, how these emotions in turn shape agency and meaning making among younger students.

First, there is the opportunity to study the longitudinal effects of whether emotionally meaningful place-based experiences lead to sustained changes in environmental attitudes and behaviours over time. As this study does not bring enough depth to these different pathways, it is difficult to draw conclusions concerning the possible long-term effects and studying how these experiences continue to shape students after the program's end would be interesting. Following students over several years could provide more insight into how early affective experiences can contribute to later transformative learning, civic participation, or environmental engagement as a whole.

There is also the possibility for further research to explore more directly the relationship between emotions and environmental agency. This study observed emotions such as awe, care, hope, and concern frequently accompanied participation and engagement within students, yet the precise mechanisms through which emotions either hinder or support future action remain underexplored. More research on this area using frameworks such as the Control-Value Theory (Pekrun, 2024) or specific eco-emotional research could help distinguish how educators could support students remaining in an “activating zone” where difficult emotions would motivate engagement rather than leading to hopelessness or disengagement.

Another area of importance within future research concerns the role of non-profit organisations and local actors within ESE. The findings of this study suggest that local volunteers, educators, scientists, and other community actors play an important role in shaping students’ emotional engagement and trust. Further research could look into how this partnership between schools and the community contributes to transformative sustainability learning and how these types of collaborations could address the limitations often found within formal schooling contexts.

Furthermore, there is the possibility for future research within this field to benefit from using alternative or complementary methodological approaches. A phenomenographic or narrative study could continue to provide more insights into how students experience environmental learning differently. Quantitative or mixed methods designs relating to

emotions, hope, environmental values, or agency could complement qualitative findings and strengthen understandings of patterns across larger populations.

Additionally, cross-cultural research would also be of value. Although this study compared two cases from different countries, the sample remained limited for various reasons. Continuing international research could explore how different countries' educational policies and culture shape sustainability learning in multiple settings.

Finally, research within this field should continue to investigate the ethical dimensions of transformative sustainability education. The challenge of encouraging environmental stewardship while avoiding fear-based approaches remains at the pedagogical core. Investigating ways in which educators can support emotionally supportive and pluralistic forms of sustainability education may be particularly important in the future of ESE research.

8. Limitations

While interpreting the findings of this study, several methodological limitations should be considered. The relatively short timeframe of the study meant that the long-term development of students' meaning-making process, sustained agency, and emerging environmental identities could not be captured over time. Although emotionally meaningful experiences and emerging forms of stewardship were observed in both cases, it is not possible to determine whether these led to lasting behavioural and value-based transformation. Transformation is inherently uncertain; it cannot be understood as a fixed, short-term outcome but as an ongoing and lifelong process in which shifts in thinking may occur immediately, after reflection, much later in life, or not at all (Biesta, 2016). The findings should therefore be understood as identifying the early "seeds" of transformative learning, rather than transformation as a whole.

The same time constraint also limited the data collection, as practical barriers meant that no student interviews could be conducted in case Blue. Had more time been available, both cases would have collected equivalent types and amounts of data. These differences were taken into account during analysis, though it remains a limitation in the comparability of the two cases.

Secondly, although both cases shared pedagogical foundations that were each place-based ESE practices, the differences of educational traditions, program implementations, geography, and culture were considerable. Notably, the Blue case was focused on ecological inquiry and future-oriented environmental learning, and the Green case was more situated within practical stewardship and repeated engagement with local nature reserves. Because of this, a direct comparison is complex. The aim is not a symmetrical comparison or creating broad

generalisations; rather, this study seeks analytical resonance across contexts by identifying overlapping themes and different meaningful contrasts. It is part of the broader reflexive practice, inherently part of transformative learning, and aims to give insight to the program designers and other programs on how to evaluate their own programs. This is in line with Sipos et al. (2008)'s intent when proposing the framework.

The findings of the study are also notably shaped by the interpretive role of each researcher. Both are experienced educators who have strong interests and backgrounds in outdoor and sustainability education, which provides a contextual sensitivity and understanding towards varying student interactions and emotions. This made them move beyond identifying emotions as facial expressions, finding them also in body language, conversations, and participation. At the same time, this positionality may have influenced the interpretation during the collection and analysis of data. It was important to have reflexive discussions between the researchers throughout the research process in order to reduce confirmation bias and strengthen the interpretative depth. Nevertheless, emotions remain an inherently difficult phenomenon to interpret through observation and conversation alone. Students' internal emotional experiences were only partly accessed through their verbal expressions, visible behaviours, and interviews.

Finally, the observational nature of the study introduced limitations regarding consistency and reliability. Language itself presents an inherent interpretive boundary: emotional expression, particularly in children, is deeply embedded in cultural and contextual nuance that resists full translation. Words, phrases, and even silences carry meaning that is shaped by local social norms and relational dynamics, dimensions that no translation, however careful, can entirely preserve. This is especially relevant when working with young children, whose emotional vocabulary is often indirect, embodied, or expressed through behaviour rather than language. As a result, some of the affective depth present in the original interactions may not have been fully captured in the analysed data. Further, the observations took place in dynamic settings, including outdoor environments involving multiple actors, changing weather settings, and varying activity structures. Although the created observation guide was used to support alignment across the cases, each data set remained context dependent.

With these limitations, the study contributes insights into the affective elements of ESE for young learners. By combining observations and interviews in addition to thematic analysis, the research is able to provide a deeper and context-sensitive understanding of how emotions, place-based experiences, and local actors come together to shape environmental learning and emerging agency within students.

9. Conclusion

Across both programs, four interrelated dimensions emerged through observations focusing on the shared emotional experiences of the students. Emotional engagement emerged through encounters with organisms, playful activities, conservation activities, and local environmental stories. Hands-on and sensory participation deepened ecological understanding. Meaning making was supported through the interaction between these emotional laden experiences, the place which affords aesthetic experiences and the social interactions between all participants. This involved connecting these experiences to sustainability concepts, with educators, local actors, and peers playing crucial roles in how students interpreted and valued what they encountered. Familiarity with the place further strengthened students' emotional connection and sense of environmental responsibility, especially when they could observe tangible impacts of their own actions.

A key difference between the two cases lies in their pedagogical emphasis and the forms of identity (trans)formation this supports. In case Green, students participated in real conservation work that directly altered local ecosystems. This provided a possible pathway to continue local stewardship in their everyday lives. In contrast, case Blue positioned students as informed investigators and future decision-makers, building systemic understanding and trust in professional environmental actors but offering fewer concrete strategies for immediate action. Both programs, however, grounded learning in strong affective engagement with local places and and the other-than-human world, and both appeared to contribute to students' emerging environmental identities.

These findings address the research questions by showing how place-based non-profit programs integrate emotional, sensory, social, and spatial dimensions of learning, and how program design choices shape the forms of environmental understanding and agency that students develop.

This thesis contributes to a broader discussion within ESE relating to the role of emotion, embodiment, and place within the learning process. Both cases displayed emotional engagement not solely functioning as a secondary outcome within learning, but as a central mechanism as students worked to interpret ecological knowledge and issues related to their lives. The data displayed students' emotional investment and overall ecological understanding to strengthen as they had different opportunities to work with trusted adults and participate in meaningful ways, especially when interacting with living organisms.

The findings of the study suggest that, as young learners are developing ways to respond constructively to environmental changes and concerns, place-based ESE programs can support this process by linking emotions with shared, meaningful action. When students expressed emotions such as worry or sadness, they were met with guidance on how they could act and collaborate in different ways, and invited to reflect on environmental degradation rather than avoid it. Both cases did this through either future-oriented environmental thinking or immediate stewardship practices, encouraging students to see themselves as active participants within different environmental communities. In doing so, the study extends existing theories on emotions related to climate change to include primary school-aged students and brings theories on social support and trust in the face of environmental challenges into the context of local actors working in non-profit organisations. It also successfully applied Control-Value Theory to place-based outdoor education, noticing how particular attention should go to the influence of contextual influences when learning moves outside the classroom.

In summary, this study demonstrates that place-based ESE programs involving local actors rooted in embodied and emotionally grounded foundations have the potential to foster the (trans)formation of identities that encompass sustainability. Although the findings are context-specific and cannot be generalized broadly, they provide insight into how ESE has the potential to move beyond knowledge transmission alone by integrating the head, heart and hands in Environmental and Sustainability Education.

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11. Appendix

Appendix 1 – Data Collection Overview

Observations Case Green

Activity	Setting	Students	Local actors	Duration
Pond-dipping, forest bath and guided walk	Nature Reserve with forest	+/- 12 students, 10-11 years old	2 volunteers from Natuurpunt + educator	135 min
Pond-dipping and guided walk	Nature Reserve with forest	+/- 15 students, 11-12 years old	1 volunteer from Natuurpunt + educator	120 min
Pond-dipping and guided walk	Nature Reserve with visitor centre	+/- 12 students, 12-13 years old, special needs school ASS	1 volunteer from the visitor centre + educator	150 min
Pond-dipping and 'living walls' activity	Abbey Park	19 students, 10-11 years old	1 volunteer from Abbey heritage conservation group + educator	120 min
Pond-dipping and guided walk	Nature Reserve with forest	15 students, 11-12 years old	1 volunteer from Natuurpunt + educator	150 min
Pond-dipping, meadow exploration and guided walk	Nature Reserve with forest	+/- 25 students, 10-11 years old	3 volunteers from Natuurpunt + educator	120 min
Nature conservation work: removing invasive trees + guided walk, water meadows, heritage + pond-dipping	Forest + nature reserve with visitor centre for water meadows	15 students, 11-12 years old	2 volunteers from Natuurpunt + educator	300 min

Official stewardship celebration + guided walk + forest activities	Nature Reserve with forest	+/- 15 students, 9-12 years old Montessori school	6 volunteers for the celebration, 3 stayed to guide activities	180 min

Interviews Case Green

Type of school	Number of students	Age of students	Total duration	Group or individual
montessori	3	9	17 min	individual
montessori	1	10	4 min	individual
montessori	2	11	15 min	individual
montessori	2	12	10 min	individual
mainstream ed	2	11 & 12	13 min	group
mainstream ed	3	11	11 min	group
mainstream ed	1	11	11 min	individual

Observations Case Blue

Activity	Setting	Students	Local actors	Duration
‘World Without Fish’ lesson- timeline activity, skit, and research project	Students' elementary school- both outside and inside activities	+/- 25 students, 11-12 years old	1 Yampatika educator, 2 classroom teachers	235 minutes
3 stations- Nature walk, Macroinvertebrates, Beavers	Yampa River State Park	+/- 45 students, 6-8 years old	3 Yampatika educators, 3 classroom teachers	285 minutes
‘World Without Fish’ final lesson- Fish hatchery visit, biodiversity games	The Finger Rock State Trout Hatchery and Rearing Unit	+/- 25 students, 11-12 years old	1 Yampatika educator, 2 fish hatchery employees	240 minutes

Appendix 2 – Interview Guide

This interview guide was translated to Dutch, and questions were adapted after observations to be more specific and to complement observations.

1. Demographics

- Can you tell me your name and age?

2. Opening question

- Can you tell me about what you did during the program?

Question 3 & 4 will be adjusted after observations to specify the activity. Question 4 is there to ask more in depth questions about emotions that arise or don't arise in question 3. The main goal is to learn more about why and how emotions arise.

3. Emotional Engagement

- How did you feel during the activity?
- What does it mean to you to do … (fill in specific observed activity).

4. Exploring different emotions

- Did you feel curious or surprised about anything?
- Was there something that made you feel excited or happy?
    - What did you do after feeling this way?
- Did you feel proud of anything you did?
- Were there moments where you felt unsure or confused?
    - What did you do after feeling this way?
- Did anything feel a bit frustrating, uncomfortable or difficult?
    - What did you do after feeling this way?
- Have you felt worried or anxious about environmental challenges in the past?
    - What makes you feel like that?
    - What do you think the future of our environment will look like?
- Did you feel like you were helping something or someone?
    - What made you feel that way?
    - What does it mean to you to take action for the environment?

5. Meaning Making

- Why do you think you were doing this activity?
    - Do you think it is important? Why or why not?
- What does learning about the environment mean for you?
    - Do you feel different learning about in class or during the program?

6. Agency

- Do you think what you did today can make a difference?
    - How?
    - Do you feel like you can help nature in other ways too?
    - Would you like to do something like this again?
- How often do you think about the effect of your actions on the environment?

7. Social dynamics

- What was it like working with your group?
- What was it like working with the volunteers?

8. Closing

- Is there anything else you want to tell me about today?
- If you could describe today in a few words, what would you say?

Appendix 3 – In-Field Observation Guide

Observation Focus	**What to look/listen for (indicators)**	**Onsite Observation Notes**	**Links to Interviews**
High-Engagement Moments	Sudden shifts in body language, gasps, focused silence, or rapid questioning/sharing		
Epistemic Emotions	Furrowed brows (confusion), "Aha!" moments, students correcting each other, and other facial expressions showing specific emotions (curiosity, surprise, confusion, or awe)		
Agency & Control	Students taking the lead, proposing a solution, or saying "I can't do this"/"this is too hard" are signs of self-efficacy		
Place-Connection	Physical contact with nature or referencing local landmarks or previous times visiting the site		
Frustration & Barriers	Sighing, disengagement, or physical withdrawal from a group task		
Collaborative	Students helping one another or discussing		
Retrospective	Linking things with past experiences		
Prospective	Linking things with future experiences		
Moments of Care	If something is mentioned of high importance (both ways- to care, or not to care)		

Appendix 4 – Consent Forms for Student Interviews

Thank you for reading the information sheet about this study. If you are happy for your child to participate, please sign the form below.

I confirm that I have read and understood the information sheet dated April 2026 and have been given the chance to ask questions about it.

I don't expect to receive any benefit or payment for my participation, and neither will my child. I understand that my participation is voluntary, and that I and/or my child are free to withdraw at any time without giving any reason and without there being any negative consequences. In addition, should my child not wish to answer any particular question or questions, they are free to decline.

I understand that my child will be informed about these rights by the researcher prior to the interview.

I understand that my and my child's responses will be kept strictly confidential. I understand that my and/or my child's name will not be linked with the research materials and will not be identified or identifiable in the report or reports that result from the research.

I agree to this interview with my child to be voice-recorded and for its contents to be transcribed. I understand that the audio recording made of this interview will be used only for the master's thesis. I understand that no other use will be made of the recording without my written permission, and that no one outside the research team will be allowed access to the original recording.

I agree for my child to take part in this interview.

I (parent/legal guardian) …….. (name) grant permission for my child …….. (name) to participate in this research.

_____	_____	_____
Your name	Date	Signature
_____	_____	_____
Researcher	Date	Signature

Appendix 5 – Information Letter Participating Schools

Dear Teacher/School Team,

My name is (name researcher). I am a primary school teacher from (place researcher) and am currently pursuing a Master's degree in Outdoor and Sustainability Education at Linköping University in Sweden. For my Master's thesis, I am conducting research during the activities of (name organisation), in which your school is participating. With this letter, I would like to inform you about the research.

Thesis Title: Emotions as a Bridge Between Learning and Action in Environmental Education

This research starts from a central question: what makes children not only learn about sustainability, but also actually act upon it? Knowledge about climate change or biodiversity is very important. However, we see that knowledge alone does not always lead to sustainable behavior. Therefore, this research focuses on an often less visible, but important factor: the role of emotions.

This research explores how children experience environmental education activities and what feelings emerge in the process. These can involve emotions such as wonder, care, pride, hope, or concern. Importantly, is how students give meaning to these experiences, and how that experience can contribute to their motivation to care for their environment. Therefore, this research focuses on what touches and moves students during a project such as this one.

Although we know that experiences in nature can have a major impact on how children view the world, it is not yet well understood exactly how that impact arises. By paying attention to what children feel and how they process their experiences, this research aims to contribute to a better understanding of how environmental education can be designed to be stronger and more meaningful, and better aligned with what motivates children.

The research is part of an international multi-case study in Belgium and the United States. In both countries, similar, practice-oriented environmental education programs are being

investigated, in which children learn actively outdoors, come into contact with their local environment, and collaborate with organizations committed to nature and sustainability.

This research stems from a deep appreciation for the work that teachers and organizations do every day. By gaining a better understanding of this practice, we hope to contribute to further strengthening environmental education in practice.

Should you have any further questions, please do not hesitate to contact us.

Thank you in advance and kind regards,
Louisa Kennedy & Stien Naets
Master's students Outdoor and Sustainability Education
Linköping University

Note: This letter was adjusted for parents to guide the consent form.

Appendix 6 – Local context

As this study will contain two case studies in different countries, a short description of the local sustainability context is provided for each.

Belgium (case Green)

Belgium is a densely populated country with relatively few natural areas. In Flanders, the northern region where this study is conducted, 32.4% of the land is primarily used for human needs, including housing, industry and agriculture. Forest takes up 11% of the total land (Departement Omgeving, 2023). Belgium still has a great diversity of biotopes, yet 7% of local species have gone extinct, and around 30% are currently on the Red List (Natuurpunt, n.d.a; Schneiders et al., 2020). Fragmentation of the landscape is one of the main causes of biodiversity loss, with 90% of its nature reserves smaller than 1 hectare, alongside pressures from climate change, pollution and urbanization (Natuurpunt, n.d.a; Schneiders et al., 2020).

There is substantial public support for stronger governmental climate action and citizen information; a recent national survey revealed that 92% of Belgians considered environmental education important, and 71% believed it should be compulsory in schools (IPSOS, 2023). This stands in contrast to recent policy decisions: the Flemish government has relatively few compulsory learning outcomes explicitly related to environmental education in its renewed curricula, and just recently decided to discontinue funding for its largest ESE support program "MOS", which helps schools work systematically on sustainability, as part of broader budget cuts (MOS Vlaanderen, 2026). On top of that, recent evaluations by the independent Federal Planning Bureau indicate that, under current policies, Belgium is not progressing fast enough to meet the Sustainable Development Goals (SDGs) in time (Federaal Planbureau, 2026).

Together, these ecological pressures, policy gaps and high levels of public support provide a fertile but challenging context for ESE in Belgium. Within this landscape, some local governments continue to support non-profit-led initiatives such as Natuurpunt's Educatief Natuurbeheer, which provides the setting for the Belgian case in this thesis.

United States (case Blue)

Steamboat Springs is found in the Northwestern area of the state of Colorado and is the setting for the United States' part of this study's case. This area is commonly defined by its vast amount of public lands and beautiful mountains, with about 75% of the landscape being covered with forests, which is notably larger than historical open grasslands (Hanberry & Seidel, 2024). Routt County has a population of around 25,000 residents, which is a demographic that has seen much of its growth driven by "amenity migration", which is where

people move to the area for its natural beauty and outdoor recreation opportunities (United States Census Bureau, 2023).

Social views in this region reflect a community that is very aware of global climate shifts and sustainability concerns, and specifically to where they live, as it is quite susceptible to these changes. Approximately 74% of Routt County residents believe that climate change is currently happening, greater than the U.S. average of 72% (Marlon et al., 2023). And 64% of the residents of the area acknowledge that climate change is largely caused by human activity (Marlon et al., 2023). This awareness is closely related to the area's place-based concerns, being that residents reported being increasingly worried about changes to the climate, specifically citing its direct impact on the local snowpack and the health of the rivers in the area, which are the lifeblood of the regional economy (Marlon et al., 2023).

The U.S. context within this study is defined by direct and observable climate impacts. All but one of the last 40 years in Colorado have been above the 20th-century temperature average, and this century has had seven of the state's ten hottest years on record. (Rocky Mountain Climate Organization, Saunders & Easley, 2018). These shifts in the climate have a notable direct impact on the Yampa River, where such rising temperatures and low flows frequently lead to river closures to protect heat-stressed fish populations (Rocky Mountain Climate Organization, Saunders & Easley, 2018).

In response to such impacts, the Steamboat Springs community has shown a large amount of agency through local governance and initiatives by local organisations and schools. The Routt County Climate Action Plan (2021) outlines different goals to work to reduce emissions and protect critical water resources, reflecting a social environment in the area where over 60% of residents support the area's climate policies (Marlon et al., 2023; Routt County, 2021). This active community engages with and creates a framework for students to transition from "worry" to "active hope" by seeing tangible local action (Routt County, 2021).

Organisations, such as Yampatika, have a goal of working together with the natural environment, the community, and with students to facilitate and learn about stewardship in response to sustainability issues on a local and global scale.